Benchmarking alignment penalties for single-cell batch correction

Kendall Reid[1,*], Genevieve Stein-O'Brien[2,†], Erhan Guven[3,†]

[1]Department of Computer Science, Johns Hopkins University; [2]Department of Neuroscience, Johns Hopkins University; [3]Johns Hopkins University Applied Physics Laboratory. *Corresponding author: kreid20@jh.edu. †These authors contributed equally.

Writer's outline. Target: *BMC Bioinformatics*, Research article; structured abstract at most 350 words; 3 to 10 keywords; no stated word, figure or reference limit; Declarations required; large language model use documented in Methods. Main text about 5,000 words plus 400 words of captions; paragraphs 100–150 words. Brackets: fill from results or still open. Background and Methods are the first author's prose with tracked corrections (`changes` package; change logs in `paper/drafts_20261022/`, `paper/drafts_20261023_pass3/` and `paper/pass4_changelog.md`. Pass 4 reframes the draft around the finished penalty-strength sweep: the benchmark at fixed strengths is not run, by the first author's decision of 2 November 2026; Pass 5 adds the replication across training seeds that replaced that benchmark, decided by the first author on the same day (change log `paper/pass5_changelog.md`)). Figures 1–6 are drawn from the sweep analysis (Figures 2–6 in Pass 10, Figure 1 in Pass 11, `paper/figure_scripts`, change logs `paper/pass10_changelog.md` and `paper/pass11_changelog.md`); Tables 2 and 3 hold sweep values (Pass 9, change log `paper/pass9_changelog.md`, number ledger `paper/pass9_numbers.csv`); Table 1 holds real design facts; the strength-range table of the planned fixed-strength benchmark, which is not run, was removed in Pass 4. Pass 13 numbers the main-text tables 1 to 3 (Table 1 the methods, Table 2 the ranking and Table 3 the placement table) and every reference to them is made by label. Pass 12 (change log `paper/pass12_changelog.md`) fixes the points of the Pass 11 rigour review; every abstract number is listed with its source in `paper/pass12_numbers.csv` and re-read by `paper/tools/check_ledger.py`. Pass 14 fills the replication across training seeds from its result (change log `paper/pass14_changelog.md`, numbers in `paper/pass14_numbers.csv`, Supplementary Table S9 generated by `paper/tools/make_replication.py`). Pass 15 fixes the points of the independent re-review and cuts the abstract (change log `paper/pass15_changelog.md`); the clean copy `paper/outline_clean.tex` is written by `paper/tools/accept_changes.py`. Pass 16 adds the family-level sign-flip sensitivity analysis to the replication result (change log `paper/pass16_changelog.md`).

Abstract

at most 350 words; Background, Results, Conclusions

1. **Background:** Single-cell batch correction often adds an alignment penalty (a discriminator, a Wasserstein-1 critic or a distance between batches) to a generative model, but penalties have rarely been compared on one backbone.
2. Question: which penalty keeps the most biology at equal batch removal in one scVI (single-cell variational inference) model, and does any beat plain scVI or standard tools?
3. Exploratory comparison on eight tasks of the single-cell integration benchmark: scVI plus the Jensen–Shannon discriminator, three Wasserstein-1 critics (reference, pooled, barycenter), maximum mean discrepancy (MMD) or the Sinkhorn divergence, each fitted at ten strengths from 0.1 to 3,000, with the decoder given the batch label or not and three seeds (100–102), beside plain scVI and standard tools. Instead of a planned fixed-strength run, the two sweep strengths bracketing matched batch removal were refitted with five new seeds (0–4).
4. **Results:** The pooled critic had the best mean rank with the batch-conditioned decoder (2.20 over five real tasks), and MMD with the unconditioned decoder (2.17 over six real tasks).
5. With the batch-conditioned decoder, the largest collapse rate on a real task ranged from 27% (12%–46%) for MMD to 73% (54%–88%) for the discriminator (nominal 95% intervals).
6. With this decoder, a penalty at one strength drawn at random exceeded plain scVI's selection score on at most 3 of six real tasks, and the best of six strengths on 4 to 6; at six tries every penalty exceeded Harmony and sysVI on 5 of six real tasks.
7. A replication across training seeds kept the sign of all four large within-pair differences (mean rank correlation 0.456, $p = 0.00049$).
8. **Conclusions:** Within these tasks, strength mattered more than penalty choice: penalties were near-tied at matched batch removal (160 of 195 within-pair differences under 0.02); a penalty exceeded batch-conditioned plain scVI mainly when several strengths were tried and scored on the same data, an optimistic gain; scANVI, trained with the cell-type labels that several biology metrics also use, exceeded every penalty on most tasks. Code and checksums: github.com/kreid415/wasserstein-critic-deconfounding.

Keywords: single-cell RNA sequencing; batch correction; adversarial learning; optimal transport; benchmarking

1 Background

Single-cell RNA-seq experiments provide a snapshot of a cell's transcriptional activity at a point in time; however, the biological differences in a dataset can be overshadowed by technical differences between experimental samples. Laboratory conditions, reagent lots, personnel and instruments introduce technical variation, which becomes a confounder when it is correlated with the biological outcome (Leek et al., 2010; Yu et al., 2024). In single-cell RNA-seq, separately processed technical replicates can differ substantially (Tung et al., 2017), and single-cell data bring new challenges for

modelling batch effects (Goh et al., 2022). This has led to a variety of batch correction methods, among them Harmony (Korsunsky et al., 2019), Scanorama (Hie et al., 2019), scVI (single-cell variational inference) (Lopez et al., 2018), its label-aware extension scANVI (Xu et al., 2021) and sysVI (Hrovatin et al., 2025). A central challenge of these methods is to balance the alignment of batches with the preservation of biological signal (Luecken et al., 2022). Under-correction results in datasets that retain substantial batch effects, while over-correction discards key biological information (Yu et al., 2024).

With the adoption of deep learning-based methods in single-cell analysis, techniques for trainable end-to-end batch correction are necessary. A notable class of deep-learning approaches, latent-variable models, aim to learn a low-dimensional representation of the RNA-seq readout that captures salient signals corresponding to biological programs. To align these latent spaces across batches, two primary methods have been adopted: decoder conditioning, which passes the batch label as an input variable to the decoder responsible for reconstructing the input RNA-seq vector from the latent variable (Lopez et al., 2018), and adversarial batch, which trains a network to predict the batch, or another confounder such as age, from each cell's latent code and penalizes the encoder whenever that prediction succeeds (Ganin et al., 2016; Dincer et al., 2020).

Adversarial methods are split between discriminator approaches, which minimize the Jensen–Shannon (JS) divergence (Goodfellow et al., 2014) between samples from different batches or confounding variables (Ganin et al., 2016; Dincer et al., 2020), and Wasserstein critic-based methods, which minimize the Wasserstein-1 (W1) distance (Arjovsky et al., 2017; Gulrajani et al., 2017) between such samples. These techniques have been applied extensively to single-cell batch correction.

iMAP trained a critic with the gradient penalty of the Wasserstein generative adversarial network (WGAN-GP), correcting every dataset towards one anchor dataset that is left unchanged (Wang et al., 2021). iDLC trains a WGAN-GP critic, with a Sinkhorn transport term added to the generator loss, to map each source batch onto a reference batch (Jiang et al., 2026); scStarCorrect also aligns every batch to one fixed reference batch, but through a domain-aware discriminator whose loss form its abstract does not state (Gu et al., 2025); Beaconet trains one Wasserstein critic per batch against the mixture of the other batches and needs no reference (Xu et al., 2024). IMGG trains a WGAN-GP generator towards a synthetic intermediate batch built from cells matched across batches (Wang et al., 2022a), ResPAN combines a Wasserstein GAN with random-walk mutual-nearest-neighbour pairing (Wang et al., 2022b), and IMAAE uses a loss of WGAN-GP form, an anchor batch and known cell labels (Wang et al., 2023). Methods built on a standard discriminator are numerous as well. Portal trains two logistic discriminators and clamps their logits so that cells scored as clearly dataset-specific give no gradient and are not forced to mix (Zhao et al., 2022); AIF pairs a real-versus-reconstructed discriminator with a batch classifier that the encoder is trained against (Monnier and Cournède, 2024); and ABC trains an autoencoder against a multi-class batch discriminator alongside cell-type classifier branches that use labels (Danino et al., 2024).

A third route avoids training an adversary network altogether by measuring the distance between the latent batches explicitly. These include the maximum mean discrepancy (MMD) (Gretton et al., 2012), and optimal transport (OT) measures (Cuturi, 2013; Genevay et al., 2018; Feydy et al., 2019). A full mathematical formulation of these approaches is available in Supplementary Note 1. Both measures have been used to align single-cell batches without an adversary. MMD-ResNet trains a residual network to match the distribution of one batch to that of another (Shaham et al., 2017), SAUCIE applies an MMD penalty against a reference batch on an internal layer (Amodio et al., 2019), trVAE places an MMD term across conditions in the first decoder layer (Lotfollahi et al., 2020), uniPort couples a variational autoencoder with minibatch unbalanced OT (Cao et al., 2022), and Li et al. added an MMD term between batches to the loss of an early scVI version (Li et al., 2022).

Despite their promise, alignment penalties have limitations that challenge their utility in practice. When batches contain different cell-type proportions, a lower bound on classification error grows as the latent distributions of the two batches are aligned more tightly (Zhao et al., 2019), and exact alignment under any divergence that is zero only for identical distributions leaves an error at least as large as the proportion gap (Wu et al., 2019). The sysVI study found that an adversarial conditional variational autoencoder and an adversarial multi-omics integration method mixed unrelated cell types when proportions were unbalanced across batches, and more so at higher adversarial strength (Hrovatin et al., 2025); dataset imbalance also changed clustering, annotation, differential expression, query mapping and trajectories across five integration methods (Maan et al., 2024).

Batch-correction methods have been compared in several large benchmarks (Tran et al., 2020; Luecken et al., 2022, 2025). These benchmarks compare complete methods, each with its own architecture, preprocessing and training settings, so they cannot show which alignment objective accounts for a method's performance. Controlled comparisons of alignment losses inside one framework are few. The extended single-cell integration benchmark (scIB-E) compared six batch losses, one of them a GAN discriminator, inside one scVI framework, with no Wasserstein, MMD or OT term (Yi et al., 2025); Li et al. compared MMD with an adversarially trained mutual-information penalty on an early scVI version and traced the Pareto front between batch removal and biology (Li et al., 2022); and the iDLC ablation removes its Sinkhorn term but has no JS or MMD arm (Jiang et al., 2026). In searches of OpenAlex, arXiv and Europe PMC

run in October 2026 and repeated in OpenAlex and Europe PMC on 5 November 2026 for works published since 1 October 2026, we found no study that compares JS, W1 and MMD or OT objectives on one stock scVI backbone. This absence is search-based, not exhaustive.

In this study we benchmark discriminator-, critic-, kernel- and optimal transport-based alignment penalties for batch correction. We compare the JS discriminator, three variants of the Wasserstein critic (reference, pooled and barycenter), MMD and the Sinkhorn divergence with standard baselines. Each penalty is added to the same scVI model (Lopez et al., 2018), with the decoder either given the batch label or not, and evaluated on eight tasks of the single-cell integration benchmark scIB (Luecken et al., 2022). The backbone and tasks are fixed, and the training settings are fixed except the adversary update steps and optimiser, which follow each method's usual setting, so differences between methods come from the alignment penalty as it is usually run. The comparison is exploratory. Every penalty is fitted across one strength range on every task and the analysis reads all of those fits. A benchmark at fixed strengths, planned to follow the sweep, was replaced by a smaller replication across training seeds: at the two sweep strengths on either side of the batch removal at which the penalties are compared (Methods), each penalty is refitted with five new seeds (0–4) on the same eight tasks, and the new fits are compared with the sweep's readings by a rule fixed before any of them was scored. The replication uses new seeds, not new data. It shows whether the sweep's readings hold when the training seed changes, and its rule tests only the sign of the within-pair differences of 0.04 or more (no precision calculation was made), so it does not test the order of penalties that sit close together.

2 Methods

To establish a fair comparison between methods, we use a fixed backbone from the well-established scVI (single-cell variational inference) architecture (Lopez et al., 2018), implemented in the scvi-tools python package (Gayoso et al., 2022). Every method uses scvi-tools' default settings and likelihood with epochs, minibatch size and latent size equal within a task, so that the alignment method and its strength are the only intended differences, apart from the adversary update steps and their optimiser, which follow each method's usual setting (Supplementary Note 2). Each adversary is trained with its standard settings: the reference, pooled and barycenter critics take five update steps per encoder step with a gradient penalty, as is usual for Wasserstein critics (Gulrajani et al., 2017), and the discriminator takes one update step, as is usual for discriminators, so the penalties are compared as they are normally run (Supplementary Note 2). We apply this across two scenarios: a conditioned setting in which the batch label is fed to the decoder, as scVI is usually run, and an unconditioned setting in which no batch label reaches the encoder or decoder, so the alignment penalty is the only correction (Figure 1a). The full hyperparameter and training details are available in Supplementary Table S2 and Supplementary Note 2.

Each method adds a fixed penalty, weighted by a fixed strength, to the normal scVI loss (Table 1). The discriminator predicts each cell's batch and the encoder minimizes scvi-tools' bounded fool loss, a Jensen–Shannon (JS) type objective (Goodfellow et al., 2014). We test three Wasserstein-1 (W1) critic-based methods(Arjovsky et al., 2017), (Gulrajani et al., 2017): reference based, pooled, and barycenter. The reference method aims to align all batches with a single reference batch, which we assign based on the highest cell-type entropy (the Shannon entropy of a batch's cell-type proportions, computed from the cell-type labels; ties go to the larger batch). The entropy rule prefers the batch whose cell types are most evenly represented, so that the target does not favour a few cell types, which a merely large batch, the other obvious choice, may do because a few cell types can dominate it. The chosen reference holds every cell type in five of the eight tasks, and 10 of 17 (lung), 14 of 19 (human–mouse immune) and 3 of 4 (simulation 2) in the others, so in those three tasks some cells of other batches have no cell of their type in the reference. The pooled critic compares each batch against the cells of all other batches pooled together. The Wasserstein barycenter approach (Cuturi and Doucet, 2014) estimates an average distribution of the batches present in each minibatch, recomputed at every step, and aligns every batch to it. The network-free approaches are the maximum mean discrepancy (MMD) of each batch to the others, with a fixed multi-scale Gaussian kernel (Gretton et al., 2012), and the debiased Sinkhorn divergence, an entropy-regularised optimal transport (OT) distance (Cuturi, 2013; Genevay et al., 2018; Feydy et al., 2019), which at the smoothing used is not an estimate of W1. For the full details of the loss equations and the training settings, see Supplementary Notes 1 and 2.

We benchmark these methods on eight integration tasks of the single-cell integration benchmark (scIB) (Luecken et al., 2022), used as distributed with their batch and cell-type labels. These consist of six real datasets from different biological settings, four of single-cell RNA sequencing (human pancreas, lung, immune cells, and joint human–mouse immune cells) and two of chromatin accessibility (assay for transposase-accessible chromatin, ATAC; mouse brain, a small and a large dataset, as gene-activity matrices), and two simulated datasets. RNA tasks use 2,000 highly variable genes and the ATAC tasks all features; counts are used as distributed. The full details of the datasets are available in Supplementary Table S1. We score each method using the scIB batch, biology, and aggregate selection scores. The batch score is the mean of five metrics: principal-component regression on batch, batch average silhouette width (ASW),

graph integration local inverse Simpson's index (iLISI), graph connectivity and the k-nearest-neighbor batch-effect test (kBET). The biology score is the mean of the task's biology metrics: normalized mutual information and adjusted Rand index between clusters and cell-type labels, cell-type ASW, isolated-label F1, isolated-label silhouette, graph cell-type LISI (cLISI), plus cell-cycle conservation on the four real RNA tasks and trajectory conservation on the two immune tasks, which have pseudotime. The mathematical formulation of these metrics is available in Supplementary Note 3. The scIB combined selection score is the weighted average $0.4\,\text{batch} + 0.6\,\text{biology}$.

Penalty-strength sweep. We fit plain scVI and the six penalties on all eight tasks at ten strengths in half-decade steps from 0.1 to 3,000 (0.1, 0.3, 1, 3, 10, 30, 100, 300, 1,000 and 3,000), with both decoders and three seeds (100–102). Two edge strengths were added in two places, 0.01 and 0.03 for the discriminator with the unconditioned decoder and 10,000 and 30,000 for the Sinkhorn divergence with the conditioned decoder, where a strength-choice rule (below) reached the end of the grid on the first three tasks fitted (the small chromatin-accessibility, immune and simulation 1 tasks). Each task therefore has 378 fits: 6 of plain scVI, 360 of the six penalties at ten strengths with three seeds and two decoders, and 12 at edge strengths; 3,024 fits in all. Nine further fits from an earlier strength-extension round, at strength 0.03 for the discriminator, the reference critic and the pooled critic on the immune task with the conditioned decoder, are also read by the analysis, which therefore uses 3,033 fits; that pair is low-signal and not ranked (Supplementary Note 4). A fit fails if its training loss or its latent becomes non-finite; failed fits are counted and left out of seed means (Supplementary Table S5). The sweep was planned as a pilot. Rules fixed before it ran were to use its seed means to choose six strengths per penalty family and decoder for a planned fixed-strength benchmark at five new seeds; that benchmark was not run, so the rules select nothing and every analysis below reads all strengths of the sweep. The comparison is therefore exploratory: it describes how each penalty behaves over its whole strength range on these tasks and tests no hypothesis fixed in advance.

Replication across training seeds. The planned fixed-strength benchmark was replaced by a smaller replication across training seeds, which asks whether the sweep's readings at matched batch removal (the next paragraph defines the matched level b^*) hold on new fits; the sweep's readings rest on three seeds per strength. For every task and decoder pair that is ranked (13 of 16; the three low-signal pairs, immune, simulation 1 and simulation 2 with the batch-conditioned decoder, are skipped) and every penalty, we took the two sweep strengths that bracket the first crossing of b^* on the sweep's seed-mean curve, with b^* fixed at its sweep value, and refitted them with five new seeds (0–4); 46 of these 78 curves cross b^* more than once, and the first crossing sets the bracket, as in the sweep reading. In one cell the first crossing lies between plain scVI and the lowest sweep strength, so plain scVI at the new seeds is its lower point. Plain scVI was refitted at the same five seeds on every task and decoder. This gives 855 fits, 775 with a penalty and 80 of plain scVI (Supplementary Note 4). The strengths were chosen from the sweep's batch scores alone, never from its biology scores, so the biology read on the new fits is not shaped by the choice. Each task's replication fits are made and scored on the same kind of machine, with the same scorer, as its sweep fits (Supplementary Note 11). The reading is the sweep's: on the means over the new seeds, biology at b^* by linear interpolation in batch score between the two bracketing strengths, walking up from plain scVI at the new seeds and taking the first crossing. If the new means do not bracket b^*, the next sweep strength outward is fitted once with five seeds; if b^* is still not bracketed, the penalty takes the bottom rank in that pair and is marked as not replicating the crossing. The sweep's readings and ranks were written down as the prediction, and the agreement rule was fixed, before any replication fit was scored (Supplementary Notes 4 and 10). The replication agrees with the sweep if two conditions hold: each of the four within-pair differences of 0.04 or more in the sweep keeps its sign on the new seeds, and the mean over the 13 ranked pairs of Kendall's rank correlation (τ_b) between the sweep ranks and the replication ranks of the six penalties is above zero by a one-sided sign-flip permutation test over pairs (all 2^{13} sign patterns, $p < 0.05$). The sign-flip test assumes that the 13 pairs' correlations are independent and symmetric about zero when the two orders are unrelated; this holds only roughly, because the two decoders of a task use the same task data and some tasks are related (two nested pairs), so its p-value is approximate. A sensitivity analysis that flips the signs per task, both decoders of a task together (all 2^8 sign patterns over the eight tasks), was added after the rule was fixed; it is reported beside the rule and is not part of it, and because it still flips the two tasks of a nested pair separately, its p-value is approximate too; a second sensitivity analysis, also added after the rule was fixed, flips the signs per task family, so that the two tasks of each nested pair and both decoders flip together (all 2^6 sign patterns over six families: the two nested pairs, pancreas, lung, simulation 1 and simulation 2, which share no data); it handles the nesting but not other dependence between tasks, so it is approximate as well. Beside the rule, and not part of it, we report the share of sweep near-ties (differences under 0.02) whose replication difference stays within ± 0.04, the rank correlation per pair, and the batch score that every replication point reached. Sweep (seeds 100–102) and replication (seeds 0–4) readings are reported separately. The rule tests only the sign of the within-pair differences of 0.04 or more, and no precision calculation was made, so the replication is not set up to separate penalties whose readings differ by less, and because it refits the same eight tasks it says nothing about other data.

Because strength has a different scale for each penalty and a stronger penalty trades biology for batch removal along a curve, as Li et al. showed for scVI (Li et al., 2022), we do not compare methods at a fixed strength. Instead we

identify, for each task and decoder, a batch score, b^*, halfway between plain scVI's mean batch score and the highest mean batch score that every one of the six penalties reaches at a sweep strength with no failed fit (Supplementary Note 5); we therefore read each penalty's biology score at b^*, a batch removal that every penalty reaches (Figure 1b). This matched level only works if the penalties remove clearly more batch effect than plain scVI. We call a task and decoder pair low-signal when that gap is under 0.02, so that b^* sits less than 0.01 above plain scVI and cannot be placed reliably; low-signal pairs are left out of the ranking, shown in a separate table. The sweep identifies which pairs are low-signal: the immune task and both simulations with the batch-conditioned decoder, where plain scVI's batch score and the highest batch score every penalty reaches were 0.638 and 0.657 for immune (gap 0.019), 0.812 and 0.819 for simulation 1 (gap 0.007) and 0.656 and 0.665 for simulation 2 (gap 0.009); no pair is low-signal with the unconditioned decoder.

The main analysis is a descriptive ranking. For each task and decoder, a penalty's curve joins plain scVI and its sweep strengths (ten, or twelve where edge strengths were fitted) by seed-mean scores over the three seeds (100–102), and its biology score is read at b^* by linear interpolation where the curve first reaches b^* (Figure 1b). For each penalty, task and decoder the uncertainty is the standard error of that reading over repeats with each of the three seeds left out in turn, holding b^* fixed and leaving the same seed out of plain scVI, and the range (lowest to highest) of the penalty's rank over the three repeats. These are reported only if at least two of the three leave-one-seed-out readings exist, and otherwise as not available; the penalty is ranked either way (Supplementary Note 5). A task and decoder pair is ranked unless it is low-signal. Within each ranked pair the six penalties are ordered by biology at b^* (rank 1 is best); tied penalties get the average of their ranks. A penalty whose curve does not reach b^* takes the bottom rank in that pair, penalties that all fail sharing the last places, following scIB, where methods that could not be run on a task were given the same rank as unintegrated data (Luecken et al., 2022). As in scIB, the mean rank over all real tasks that are not low-signal is reported for each decoder; simulations are ranked and reported separately, and low-signal pairs are shown separately and left out of the ranking (Supplementary Note 6). We report no significance tests: ranks and their uncertainty describe the data. Each penalty's selection score is also compared with plain scVI's, at one strength not tuned on the task and as the expected best of k tries (below). The ranking rules were written for the planned fixed-strength benchmark and are applied to the sweep with the changes listed in Supplementary Note 10: the three tasks fitted first had already been read by the strength-choice rules, whereas no score of the other five tasks existed when the ranking rules were written (22–23 October 2026).

We place the penalties against plain scVI and against standard tools: scvi-tools' own adversarial classifier (its built-in batch classifier, run with the conditioned decoder only), Harmony (Korsunsky et al., 2019), Scanorama (Hie et al., 2019), sysVI (Hrovatin et al., 2025), scANVI, which uses the cell-type labels (Xu et al., 2021), and principal-component analysis (PCA) as the uncorrected reference. We compare all of these approaches at equal numbers of tries, allowing for the outcome that no alignment penalty improves on plain scVI. Harmony, Scanorama and sysVI run at six settings each, scANVI and PCA at one (settings, seeds and fit counts in Supplementary Table S4 and Supplementary Note 8). A try is one setting of a method, fitted and scored. The scvi-tools adversary, scANVI and sysVI are fitted with five seeds (0–4), and Harmony, Scanorama and PCA once per setting, all on every task: 448 fits, including three sensitivity settings (Harmony at 50 principal components, Scanorama at 100 dimensions and PCA at 50 components). Because the penalties and the tools have different numbers of settings, placement compares methods at equal numbers of tries, by the expected best selection score among k settings drawn at random, and is descriptive. Each penalty is also scored at one strength not tuned on that task, the strength with the best mean selection score on the real tasks of the other task families, shown beside the task's own best strength as an optimistic bound (Supplementary Note 7). Harmony, Scanorama and sysVI get their setting by the same rule: the setting with the best mean selection score on the real tasks of the other families, with the task's own best setting in parentheses as the optimistic bound (Supplementary Note 8). scANVI and PCA run at one setting, so they involve no choice.

For each penalty we also report how often it collapses and how narrow its range of good strengths is, together with its hypervolume, its share of non-dominated strengths, its local versus global batch removal and label transfer; all six views are descriptive, and each rule is given in full in Supplementary Note 7. A fit collapses if it fails or its biology score is at least 0.10 below the same seed's plain scVI; the collapse rate is the collapsed share of a penalty's fits in a task and decoder, with exact binomial (Clopper–Pearson) 95% intervals (Clopper and Pearson, 1934), which are nominal because the fits of one seed are judged against one plain-scVI threshold and are not independent (Discussion). The good-strength width is the span, in decades, of the strengths whose mean selection score is at least the lower 95% confidence bound of the best strength's mean. The hypervolume (Zitzler and Thiele, 1998), which Li et al. used to compare trade-off fronts (Li et al., 2022), is the part of the batch-score by biology-score plane that a penalty's sweep points reach beyond a reference point at plain scVI's batch score and 0.10 below its biology score, divided by the batch gap that sets b^* so that tasks share one scale. The share of non-dominated strengths, also from Li et al., is the fraction of a penalty's strengths that none of its other strengths beats on both the batch and the biology score.

Local removal is the mean of the three neighbourhood batch metrics (iLISI, kBET and graph connectivity) and global removal the mean of the two whole-embedding metrics (principal-component regression and batch ASW), so

that at b^* more local removal means less global; we report each penalty's local minus global reading at its crossing of b^*. Label transfer holds each batch out in turn and predicts its cell types from the other batches with a 15-nearest-neighbour classifier, on each penalty's embedding at the strength whose batch score is nearest b^*, never chosen by accuracy (Supplementary Note 9).

A direct biology check reads four per-cell-type quantities on the saved embeddings, with no new fits: neighbourhood purity, the share of a cell's 15 nearest neighbours in the latent space that carry its own cell-type label, averaged within each type; the same purity and the label-transfer recall for rare types, those with under 1% of a task's cells; marker-gene preservation, the Jaccard overlap between each type's top 20 markers from the expression as distributed and from the expression averaged over the 15 neighbours, on the four RNA tasks; and the label-transfer confusion table. Penalties are read at the label-transfer strength and every baseline at each of its settings, and each value is reported as a difference from plain scVI. The check is exploratory: its rules were written after the sweep scores had been read and before any of its values was computed (Supplementary Note 12).

Large language model tools were used for writing and checking analysis code, running and checking computational jobs, and drafting and editing text. The authors checked all outputs and take responsibility for the content [*to be decided: first author to confirm tools, versions and roles*].

Results

1,500 words; 8 paragraphs plus one for the replication across training seeds and one for the direct biology check

Res-1. We first report which fits completed and where batch removal could be matched.
All 3,033 sweep fits completed, with 0 failed fits across the penalties and decoders (Supplementary Table S5). The analysis includes 387 fits on the immune task and 378 on each other task; the immune total includes additional strengths for the discriminator, the reference critic and the pooled critic with the batch-conditioned decoder. Of the 448 baseline fits, 447 were scored and 0 failed: the scvi-tools adversary 40 of 40, scANVI 40 of 40, sysVI 240 of 240, Harmony 56 of 56, Scanorama 55 of 56 and PCA 16 of 16. The remaining fit, Scanorama with 80 neighbours on the large chromatin-accessibility task, could not be fitted (it ran out of memory, and a retry stopped at a 32-bit sparse-matrix index limit); it is reported as an incomplete setting, not as a failed fit, so on that task Scanorama is placed with five of its six settings (Supplementary Table S5). The low-signal pairs were the immune task and both simulations with the batch-conditioned decoder: plain scVI and the highest common batch scores were 0.638 and 0.657 (gap 0.019) for the immune task; 0.812 and 0.819 (gap 0.007) for simulation 1; 0.656 and 0.665 (gap 0.009) for simulation 2. Every penalty reached b^* in every task and decoder pair, including the low-signal pairs; none received a bottom rank for a missing full-seed reading. Low-signal pairs are reported separately under the Methods rule (Supplementary Note 5).

Res-2. Penalties are ranked by the biology they keep at matched batch removal (Figure 2, Table 2).
With the batch-conditioned decoder, the ranking rested on five real tasks. Mean ranks (range across those tasks) were the pooled critic 2.20 (1–3); the Sinkhorn divergence 2.40 (1–5); the barycenter critic 3.00 (1–5); the reference critic 3.80 (1–5); MMD 4.60 (2–6); the discriminator 5.00 (3–6). With the unconditioned decoder, all six real tasks contributed: MMD 2.17 (1–6); the reference critic 3.33 (1–6); the pooled critic 3.33 (2–5); the discriminator 3.50 (1–5); the barycenter critic 4.00 (1–6); the Sinkhorn divergence 4.67 (3–6). The Sinkhorn divergence ranked first most often with the batch-conditioned decoder, on the human–mouse immune and lung tasks; the discriminator ranked last on the small chromatin-accessibility, lung and pancreas tasks. With the unconditioned decoder, MMD ranked first most often, on the large chromatin-accessibility, human–mouse immune and pancreas tasks; the reference critic ranked last on the small chromatin-accessibility and human–mouse immune tasks, and the Sinkhorn divergence on the large chromatin-accessibility and lung tasks.
Neighbouring biology readings often differed by less than at least one of their reported leave-one-seed-out standard errors (Methods; Supplementary Table S6). For example, with the batch-conditioned decoder on the small chromatin-accessibility task, the pooled critic and the Sinkhorn divergence read 0.632 and 0.631, with standard errors 0.007 and 0.004, respectively. With the unconditioned decoder on the large chromatin-accessibility task, MMD and the reference critic read 0.622 and 0.621, with standard errors 0.010 and 0.006. Some gaps exceeded both neighbouring standard errors: on the small chromatin-accessibility task with the batch-conditioned decoder, MMD exceeded the reference critic by 0.017 (standard errors 0.007 and 0.003). The simulations were reported separately (Supplementary Figure S1): with the unconditioned decoder, the order from highest to lowest biology was the discriminator, the reference critic, MMD, the Sinkhorn divergence, the pooled critic and the barycenter critic in simulation 1, and the discriminator, the pooled critic, MMD, the reference critic, the barycenter critic and the Sinkhorn divergence in simulation 2. Neither simulation was ranked with the batch-conditioned decoder because both pairs were low-signal.

Res-2b. The replication across training seeds reports whether the sweep's readings at b^* hold on new fits (Supplementary Table S9).

- Rule: the sweep's readings and ranks at b^* in the 13 ranked pairs are the prediction; the replication agrees with the sweep if each of the four large sweep gaps (0.04 or more within a pair) keeps its sign and the mean Kendall's τ_b over

the 13 pairs is above zero by the sign-flip permutation test; everything else in this paragraph is descriptive (Methods; Supplementary Notes 4 and 10).

- All 895 replication fits were scored and 0 failed: the 855 planned fits and 40 fits of the extension step, which refitted the next sweep strength once, with seeds 0–4, for the 8 readings whose new-seed means did not bracket b^* (Supplementary Note 11). Plain scVI at seeds 0–4 reached batch scores within -0.006 to 0.005 of the sweep's plain scVI in the 13 ranked pairs. Before the extension step 70 of the 78 readings bracketed b^*; after it 77 did. One reading, MMD on the pancreas task with the batch-conditioned decoder, did not replicate the crossing (its highest new-seed mean batch score was 0.674, against $b^* = 0.684$) and takes the bottom place in its pair.
- Biology at b^* on the new seeds differed from the sweep reading by a median of 0.004 and at most 0.024 (absolute differences over the 77 readings that reach b^*); the mean absolute difference per pair ranged from 0.002 to 0.011. In 7 readings the new-seed mean at the lower refitted strength had already reached b^*, so the reading interpolates from plain scVI at seeds 0–4. The first-ranked penalty was the same as in the sweep in 6 of 13 pairs; 29 of the 78 penalties kept their sweep rank and 56 moved by at most one place. Supplementary Table S9 gives every reading with the batch and biology scores of its two bracketing points.
- Each of the four large sweep gaps kept its sign on the new seeds (4 of 4); the closest to flipping was the barycenter critic minus the reference critic on the small chromatin-accessibility task with the unconditioned decoder (sweep +0.041, replication +0.013). Kendall's τ_b ranged from -0.067 to 0.867 over the 13 pairs (11 positive), with mean 0.456 and one-sided sign-flip $p = 0.00049$ (4 of the 2^{13} sign patterns reach the observed mean). The agreement rule is met. Before the extension step that gap could not be read, because the barycenter critic's new-seed mean had not reached b^*; the other 3 gaps kept their sign and the mean τ_b was 0.426 ($p = 0.00049$), so the rule was met only once the step it prescribes had been run.
- Sensitivity analysis, not part of the rule: with the signs flipped per task, both decoders of a task together (all 2^8 patterns over the eight tasks), $p = 0.0039$. Only the observed pattern reaches the observed mean (1 of 256), because the summed τ_b is positive in all 8 tasks, so this is the smallest p that eight tasks allow. It keeps the two decoders of a task together but still flips the two tasks of each nested pair separately, so it is approximate as well (Methods). With the signs flipped per task family (the two tasks of each nested pair and both decoders together; all 2^6 patterns over six families), $p = 0.0156$. Only the observed pattern reaches the observed mean (1 of 64), because the summed τ_b is positive in all 6 families, so this is the smallest p that six families allow; it was defined after the rule and is approximate as well (Methods).
- Of the 160 sweep near-ties (differences under 0.02), 155 kept a replication difference within ± 0.04 (127 before the extension step). Near-tied penalties stayed close, but the rule tests only the sign of the large gaps, so their order is not tested by it (Methods). The slope-adjusted reading, a sensitivity analysis whose formula was defined after the rule was fixed (the rule names it without one), kept the sign of 4 of the 4 large gaps, with mean τ_b 0.374 and $p = 0.00073$.

Res-3. Figure 3 shows each method's trade-off between batch removal and biology in every real task.

The real-task curves overlapped near plain scVI and around many matched readings, but differed more at stronger penalties (Figure 3; curve construction in Supplementary Note 5). With the batch-conditioned decoder, the lung and pancreas curves were close near b^*, whereas the discriminator lost more biology at stronger settings. On the large chromatin-accessibility task, the barycenter critic reached a lower maximum batch score, 0.674, than the reference critic, 0.815. This shorter batch-removal range did not reflect failed fits or failure to reach b^*. At the strongest fitted setting on lung with the batch-conditioned decoder, the barycenter critic ended at batch and biology scores of 0.466 and 0.397, below plain scVI on both scores. With the unconditioned decoder, the chromatin-accessibility curves lost biology as batch removal increased, whereas on pancreas several penalties increased both scores over part of their strength range. Thus, a curve could turn back or lose biology after reaching the matched level; no real-task curve ended early because of failed fits.

Res-4. Hypervolume, share of non-dominated strengths and collapse rate give three further views of each penalty (Figure 4a,b, Supplementary Table S8).

The secondary summaries varied across real tasks (Figure 4; Supplementary Table S8; definitions in Supplementary Note 7). Hypervolume ranges with the batch-conditioned and unconditioned decoders, respectively, were the discriminator 0.029–0.255 and 0.060–0.386; the reference critic 0.022–0.269 and 0.038–0.383; the pooled critic 0.047–0.344 and 0.074–0.440; the barycenter critic 0.055–0.108 and 0.072–0.128; MMD 0.039–0.178 and 0.089–0.390; the Sinkhorn divergence 0.060–0.155 and 0.085–0.192. The corresponding ranges of the share of non-dominated strengths were the discriminator 0.400–0.545 and 0.083–0.583; the reference critic 0.364–0.600 and 0.100–0.900; the pooled critic 0.364–0.600 and 0.200–0.700; the barycenter critic 0.200–0.600 and 0.200–0.600; MMD 0.300–0.600 and 0.300–0.700; the Sinkhorn divergence 0.250–0.500 and 0.300–0.600. These shares and collapse summaries include the low-signal immune pair. The largest real-task collapse rate for each penalty, with its exact nominal 95% interval, was as follows with the batch-conditioned decoder: the discriminator 73% (54%–88%; the human–mouse immune task); the reference critic 57% (37%–75%;

the small chromatin-accessibility task); the pooled critic 47% (28%–66%; the small chromatin-accessibility task); the barycenter critic 33% (17%–53%; the small chromatin-accessibility task); MMD 27% (12%–46%; the small chromatin-accessibility task); the Sinkhorn divergence 42% (26%–59%; the small chromatin-accessibility task). With the unconditioned decoder, the corresponding maxima were the discriminator 61% (43%–77%; the small chromatin-accessibility task); the reference critic 57% (37%–75%; the small chromatin-accessibility task); the pooled critic 57% (37%–75%; the large chromatin-accessibility task); the barycenter critic 30% (15%–49%; the large chromatin-accessibility task); MMD 13% (4%–31%; the large chromatin-accessibility task); the Sinkhorn divergence 30% (15%–49%; the small chromatin-accessibility task). All recorded collapses reflected loss of biology rather than failed fits. MMD had lower collapse rates than the discriminator on every real task with the batch-conditioned decoder, although it ranked above the discriminator on only some of the ranked tasks.

Res-5. We then report whether each penalty removes batch effects more locally or more globally, and the label-transfer check (Figure 4c; label transfer in Supplementary Figure S3).
At b^*, every penalty had more global than local batch removal on the ranked real tasks except the large chromatin-accessibility task, where the local-minus-global reading was positive (Figure 4c; Methods and Supplementary Note 7). On that task, readings for the discriminator, the reference critic, the pooled critic, the barycenter critic, MMD and the Sinkhorn divergence were 0.043, 0.041, 0.043, 0.039, 0.036, 0.039 with the batch-conditioned decoder and 0.105, 0.109, 0.111, 0.113, 0.106, 0.091 with the unconditioned decoder. The ordering was task-dependent: the discriminator had the highest local-minus-global reading on lung with the batch-conditioned decoder (-0.356), whereas the Sinkhorn divergence had the highest on lung with the unconditioned decoder (-0.289). In the same penalty order, mean label-transfer macro F1 over the ranked real tasks was 0.752, 0.790, 0.771, 0.796, 0.799, 0.769 with the batch-conditioned decoder and 0.726, 0.801, 0.818, 0.815, 0.811, 0.783 with the unconditioned decoder (Supplementary Figure S3; Supplementary Note 9). These orders differed from the biology ranking at b^*. Label-transfer embeddings were not exactly matched: for example, on the small chromatin-accessibility task with the batch-conditioned decoder, the pooled critic and the Sinkhorn divergence had macro F1 scores of 0.840 and 0.851, at batch-score distances of 0.024 and 0.010 from b^*, respectively. Along the full strength curves, label transfer also deteriorated at some strong settings: the barycenter critic ended at 0.360 on lung with the batch-conditioned decoder.

Res-5b. A direct biology check reads purity, rare types, marker genes and label-transfer confusion per cell type (Supplementary Table S10).
This check is exploratory and descriptive: it was defined after the sweep scores had been read, uses 783 saved fits (penalties at the label-transfer strength, seeds 100–102; plain scVI; every scored baseline setting) and runs no test (Methods; Supplementary Note 12). Over the six real tasks, neighbourhood purity per cell type differed from plain scVI with the same decoder and seed by -0.052 for the discriminator with the batch-conditioned decoder and -0.077 with the unconditioned decoder, and by -0.022 and -0.033 for the Sinkhorn divergence; the reference, pooled and barycenter critics and MMD stayed between -0.019 and $+0.001$ with either decoder. Harmony, averaged over its 7 settings, differed from batch-conditioned plain scVI by -0.141, and scANVI, which is trained with the labels, by $+0.115$. Rare types (present on the pancreas, lung, immune and human–mouse immune tasks) lost purity in 23 of 48 task, penalty and decoder cells, most with the discriminator (-0.107 with the batch-conditioned decoder). The largest drop, -0.406, was for the 46 ionocytes of the lung task with the batch-conditioned discriminator at strength 3,000: their label-transfer recall was 0.326 (seed SD 0.407) against 0.797 for plain scVI, and a share of 0.232 of them was predicted as dendritic cells. Over all cell types, label-transfer recall differed from plain scVI by -0.040 and -0.085 for the discriminator with the two decoders and by -0.024 to $+0.003$ for the reference, pooled and barycenter critics and MMD. Marker genes were kept alike: the Jaccard overlap of the top 20 markers was 0.390 to 0.399 for every penalty and plain scVI, with differences from plain scVI of at most 0.006, whereas Harmony (-0.031) and PCA (-0.028) were lower. The direct readouts single out the discriminator, which also had the worst mean rank with the batch-conditioned decoder and the lowest mean label-transfer F1 with both decoders (Res-5); with the batch-conditioned decoder its loss of purity was larger for rare types than over all types.

Res-6. The range of good penalty strengths is reported for every method (Figure 5).
Across the real tasks, including the low-signal immune pair, good-strength widths with the batch-conditioned and unconditioned decoders, respectively, ranged as follows: 0.0–4.5 and 0.0–3.5 decades for the discriminator; 0.0–3.5 and 1.0–4.0 decades for the reference critic; 0.0–2.0 and 0.5–4.5 decades for the pooled critic; 0.0–3.5 and 0.0–2.5 decades for the barycenter critic; 0.0–4.5 and 0.0–3.5 decades for MMD; 2.0–5.5 and 1.0–4.5 decades for the Sinkhorn divergence (Figure 5; definition in Supplementary Note 7). The Sinkhorn divergence had a nonzero width on every real task with either decoder. Each other penalty had a zero-width window on at least one real task and decoder pair; a zero width indicates a single qualifying strength, not a missing result. For example, MMD had widths of 4.5 decades on each chromatin-accessibility task with the batch-conditioned decoder but 0.0 on the human–mouse immune task. Every real-task pair had a defined window. The best strength lay at a grid edge for some penalties: on pancreas with the batch-conditioned decoder, MMD was best at the largest fitted strength, 3,000; on the large chromatin-accessibility

task with the unconditioned decoder, the Sinkhorn divergence was best at the smallest, 0.1. Some windows contained gaps between qualifying strengths, so their reported span need not represent a continuous range of good strengths.

Res-7. Single-strength tables place each method against plain scVI and the standard tools (Table 3, Figure 6b).

- Rule: each penalty, and each of Harmony, Scanorama and sysVI at one of its settings, is shown at a strength or setting not tuned on that task, the one with the best mean selection score on real tasks of other families (related tasks never choose for each other), beside the task's own best strength as an optimistic bound; descriptive (Supplementary Notes 7 and 8, Table S4).
- At the strength chosen on real tasks of other families, the penalties with the batch-conditioned decoder exceeded plain scVI's selection score on 3 (discriminator), 4 (reference critic), 5 (pooled critic), 5 (barycenter critic), 4 (MMD), 4 (Sinkhorn divergence) of the six real tasks, with differences from plain scVI from −0.023 to +0.045 (median +0.006; Table 3); 11 of these 36 differences were smaller than 0.005 in size (for example −0.0004 for the Sinkhorn divergence on pancreas, which Table 3 prints as a tie), against seed standard deviations of the penalty's selection score of 0.001 to 0.016 in those cells, so the counts mix small gains, small losses and near-ties; at each task's own best strength, the optimistic bound, they did so on 5 to 6 of six. At their transferred settings, Harmony, Scanorama and sysVI exceeded batch-conditioned plain scVI on 1, 0 and 1 of six real tasks. With the unconditioned decoder, whose plain scVI receives no batch information, the penalties at the transferred strength exceeded plain scVI on 2 to 6 of six (Supplementary Table S7). The transferred strengths with the batch-conditioned decoder were discriminator 0.1–0.3; reference critic 0.3–3; pooled critic 1; barycenter critic 0.3–3; MMD 30; Sinkhorn divergence 1–100.

Res-8. The comparison at equal numbers of tries reports whether any penalty improves on plain scVI (Figure 6a; per-task curves in Supplementary Figure S6).

- Rule: each method gets the expected best selection score among k of its settings drawn at random, k = 1–6, the number of settings of Harmony, Scanorama and sysVI (comparisons at equal tries stop at six, although the penalties have ten or twelve sweep strengths), so methods with more settings gain no advantage; optimistic because the best setting is chosen and scored on the same data; scANVI has its own stratum (Supplementary Note 7).
- With the batch-conditioned decoder, a penalty's expected best selection score at one try exceeded plain scVI's on 0, 2, 2, 0, 3, 1 of the six real tasks (in the order discriminator, reference, pooled and barycenter critics, MMD, Sinkhorn divergence), 8 of 36 task–penalty cells, and at six tries on 4, 5, 5, 5, 5, 6 (30 of 36). Averaged over the six real tasks, plain scVI scored 0.648; the penalties ranged from 0.609 to 0.653 at one try and from 0.661 to 0.666 at six (Figure 6a), and their real-task means first exceeded plain scVI's at 4, 3, 2, 2, 1, 2 tries in the same order. Against the standard tools at six tries, every penalty exceeded Harmony on 5 of six real tasks, sysVI on 5 of six and Scanorama on all 5 tasks where its six settings exist; at one try the counts were Harmony 4 to 5, sysVI 4 to 5 and Scanorama 6 of six. The scvi-tools adversary (mean 0.651) was exceeded at one try on 0 to 4 of six tasks and at six tries on 5 to 6. scANVI, which uses the cell-type labels (mean 0.700), was exceeded by no penalty at one try with either decoder; at six tries with the batch-conditioned decoder every penalty exceeded it on pancreas only, by 0.002 to 0.010, and with the unconditioned decoder on no task. With the unconditioned decoder the reference is plain scVI without batch information (mean 0.578), which the penalties exceeded at one try on 1 to 5 of six real tasks and at six tries on 5 to 6. Every best-of-k value is optimistic for methods with more than one setting, because the best setting is picked and scored on the same data (Supplementary Note 7).
- Descriptive; best-of-k labelled optimistic.

Discussion

700 words; 5 paragraphs

Disc-1. The penalty with the best mean rank depended on the decoder: the pooled critic ranked highest with the batch-conditioned decoder, followed closely by the Sinkhorn divergence, whereas MMD ranked highest with the unconditioned decoder. The task-level orders varied, and many neighbouring biology readings were within a leave-one-seed-out standard error. These descriptive rankings therefore do not establish a consistent ordering across tasks. The secondary summaries also gave different orders: MMD had relatively low collapse rates despite ranking near the bottom with the batch-conditioned decoder, and the local-versus-global readings did not give a consistent advantage to one penalty. This exploratory comparison on eight tasks with three seeds (100–102) supports a decoder-specific description of the sweep; the replication across training seeds, on new seeds of the same tasks, kept the four large within-pair gaps and agreed with the pair orders in part (Res-2b).

- The replication across training seeds met its agreement rule: the four large within-pair gaps kept their sign, and the sweep and replication orders were positively correlated (mean τ_b 0.456, $p = 0.00049$; $p = 0.0039$ with signs flipped per task and $p = 0.0156$ per task family; all three values are approximate, because related tasks are not independent). The orders did not hold in detail: the first-ranked penalty changed in 7 of 13 pairs, τ_b was slightly negative in 2 pairs (pancreas and the human–mouse immune task, both with the batch-conditioned decoder; $\tau_b = -0.067$), and MMD on pancreas with the batch-conditioned decoder did not replicate its crossing. Because the replication refits

the same tasks with new seeds, not new data, and its rule tests only the sign of gaps of 0.04 or more, it supports the large gaps and the broad order, not the order of near-tied penalties.

Disc-2. The secondary analyses suggest that the penalties differ partly in how much biology they lose as strength increases and in how broadly useful strengths are distributed within the fitted range. The discriminator often lost biology beyond the matched level, while MMD generally had lower collapse rates; nevertheless, lower collapse did not ensure a higher biology rank at b^*. Hypervolume and the share of non-dominated strengths describe different aspects of the curves, so their orders need not agree with a ranking at a single batch score. Likewise, label-transfer scores did not follow the matched-biology ranking, and their batch-score distances from b^* limit direct comparisons. Local-versus-global differences varied by task, suggesting no common local-removal explanation for the ranking. These patterns describe each penalty as it is usually trained, with each adversary's update steps and optimiser at its standard setting (Methods).

Disc-3. For users of scVI, the sweep points to the penalty strength, more than the choice among penalties, as the setting that matters. At matched batch removal the six penalties were near-tied: 160 of the 195 within-pair differences in biology over the 13 ranked pairs were under 0.02, and 4 were 0.04 or more. With the batch-conditioned decoder, going from one to six tries raised each penalty's mean expected best selection score over the real tasks by 0.012 to 0.053, whereas at six tries the six penalties' means spanned only 0.661 to 0.666. Added to scVI as it is usually run, with the batch-conditioned decoder, a penalty at one strength drawn at random rarely beat plain scVI (8 of 36 task–penalty cells); the best of six strengths did in 30 of 36, by small margins (real-task means 0.661–0.666 against 0.648).

- When to add a penalty: when several strengths can be fitted and scored on the task, because the real-task mean of the best of k first exceeded batch-conditioned plain scVI at 1 (MMD) to 4 (discriminator) tries. Strength range: the strengths chosen on other task families with the batch-conditioned decoder lay between 0.1 and 100. Standard tools: at equal tries the penalties exceeded Harmony, Scanorama and sysVI on most real tasks, but not scANVI, which uses cell-type labels and was exceeded only on pancreas at six tries. These are optimistic best-of-k readings on eight scIB tasks.

Disc-4. The design limits what the data can show.

- Six real tasks in four families, two of them nested pairs (human–mouse immune contains all immune cells; large ATAC contains small); related tasks count as separate tasks in the mean rank, and the ranking is descriptive. A penalty whose curve does not reach b^* is ranked last in that pair, so a missing reading counts as the worst result.
- The comparison is exploratory. The strength grid was fixed in advance and the sweep reports all ten strengths, none left out (the strengths of the single-strength tables and of the replication were then chosen from sweep scores, as Methods describes); the planned fixed-strength benchmark that was to follow the sweep (six strengths per family, five new seeds) was replaced by the smaller replication across training seeds. The sweep results describe the sweep and test no hypothesis fixed in advance, and the ranking rules were applied to the same fits that the strength-choice rules had already read. Three seeds per strength: the standard errors in Supplementary Table S6 show how far apart neighbouring ranks are. The replication checks the sweep's readings at b^* on new fits against a prediction and an agreement rule fixed before any replication score was read, but it refits the same eight tasks with new training seeds, not new data, so it says nothing about other datasets, batch structures or preprocessing. Its rule tests only the sign of the within-pair differences of 0.04 or more, and no precision calculation was made, so it does not separate penalties whose readings differ by less. Its strengths and b^* were set from the sweep's batch scores after the sweep had been read, so it is a replication of the sweep's readings, not a test fixed before the sweep.
- All fits use one backbone, scVI, with one fixed architecture and latent size (10 latent dimensions, one hidden layer of 128 units), so the ranking may not hold for other backbones, architectures or latent sizes. The eight tasks are of the size of the single-cell integration benchmark, not of an atlas, so the ranking may not hold for integrations of that scale. The replication across training seeds refits the same eight tasks, so it tests the sweep's readings against the training seed and not against new datasets. The biology checks, including label transfer, are exploratory: they were defined after the sweep scores had been read.
- The strength sweep is what Li et al. call scalarization: one penalty is run at a series of fixed strengths and both scores are read at each (Li et al., 2022). The ranking therefore describes the penalties as users run them, not the best trade-off front that a multi-objective solver could reach for each penalty. Li et al. also used an adversarially trained mutual-information penalty (MINE); it is not among the six penalties and was not tested here.
- One batch variable per task; ATAC as gene-activity matrices; non-integer counts as distributed; scoring includes training cells; Sinkhorn not W1; reference choice uses labels.
- The collapse-rate intervals are nominal exact binomial 95% intervals. They treat a penalty's fits in a task and decoder (30; 33 for the discriminator, the reference critic and the pooled critic on the immune task with the batch-conditioned decoder, which have additional strengths; 36 where edge strengths were fitted) as independent, although each seed's fits are judged against the same plain-scVI threshold and can collapse together, so their actual coverage may differ from 95%. An illustrative simulation of this dependence is in Supplementary Note 7; it does not bound the coverage

of the intervals reported here.

Disc-5. Several factors are outside the scope of this comparison.

- Not varied or tested: architecture, Kullback–Leibler (KL) warm-up (incomplete on shorter tasks, penalty from start), reference batch, batch number; composition shift read from strength curves, not a dose-response test.

Conclusions

about 150 words; 1 paragraph

- On eight tasks of the single-cell integration benchmark, six alignment penalties added to one scVI model kept similar biology at matched batch removal: the best mean rank was the pooled critic's with the batch-conditioned decoder (2.20) and MMD's with the unconditioned decoder (2.17), but 160 of the 195 within-pair differences were under 0.02, so the ranking does not separate most penalties. At equal numbers of tries with the batch-conditioned decoder, the penalties exceeded Harmony, Scanorama and sysVI on most real tasks and plain scVI mainly when several strengths were tried and scored on the same data; scANVI, which uses cell-type labels, was exceeded only on pancreas.
- Observations for scVI users, from the scores on these eight tasks and not tested on independent data: (i) the strength mattered more than the choice of penalty, and plain scVI with the batch-conditioned decoder remained a strong candidate, since a penalty at one strength drawn at random beat it in 8 of 36 task–penalty cells and the best of six strengths in 30, a count that is optimistic because the strengths were chosen and scored on the same data; (ii) the strengths chosen on other task families with that decoder lay between 0.1 and 100 across penalties (Results); (iii) the biology was worth checking directly at the chosen strength, because the discriminator lost the most neighbourhood purity in the direct biology check (-0.052 against plain scVI, -0.107 for rare types, batch-conditioned decoder); (iv) when cell-type labels were available, scANVI scored higher than every penalty on most tasks, but it is trained with the labels that several biology metrics also use, so its biology scores are not label-free.

Declarations

eight BMC subheadings, each filled or marked not applicable

Ethics approval and consent to participate. Only publicly available, previously published data are analysed (the tasks of the single-cell integration benchmark). [*to be decided: confirm wording*]

Consent for publication. Not applicable.

Availability of data and materials. Code and data access.

- Repository github.com/kreid415/wasserstein-critic-deconfounding; fitting and scoring code at commit 10a1ca5, sweep analysis at commit 8595a16, biology check at commit c4a793f, figure scripts and their inputs in `paper/figure_scripts` and `paper/figure_data`; [*to be decided: release tag and commit of the submitted version*]; checksums in supplement.
- The eight task files are the integration-task datasets of the single-cell integration benchmark, figshare article 12420968, version 8 (doi:10.6084/m9.figshare.12420968.v8); file checksums in Supplementary Table S1.

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Supplementary material

Supplementary Notes 1–11, Tables S1–S8 and S9, Supplementary Note 12 and Table S10, Figures S1–S6

This document contains Supplementary Note 1, Tables S6, S7, S9 and S10 and Figures S1–S6; the other notes and tables listed below are described here and are not yet written.

Supplementary Notes

- Supplementary Note 1, model and loss equations: notation, with the reconstruction term written per gene g as $p_{\mathrm{ZINB}}(x_g \mid \mu_g, \theta_g, \pi_g)$ (mean, dispersion, dropout) and the discriminator loss as $\min_h \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \mathcal{L}_{\mathrm{CE}}(h(u_i), s_i)$ on the posterior mean $u_i = \mu_i$ of cell i in minibatch $\mathcal{B}$, for batch label s_i; scVI loss (zero-inflated negative binomial reconstruction, Kullback–Leibler (KL) term and KL weight schedule); discriminator fool loss as in scvi-tools (Jensen–Shannon type); the three Wasserstein-1 critics with gradient penalty weight 10: reference (each batch against the reference batch, chosen as the batch with the highest cell-type entropy, which uses labels), pooled (each batch

against all others), barycenter (each batch against the minibatch Wasserstein barycenter, an average distribution); barycenter target construction and solver (Wasserstein-2, at most 10 iterations, threshold 10^{-7}); maximum mean discrepancy (MMD) with multi-scale Gaussian kernel (the squared MMD of each batch against the cells of all other batches in the minibatch, as a biased V-statistic, averaged over the batches present; the kernel is the mean of five Gaussian kernels $\exp(-\|z - z'\|^2/2\sigma^2)$ with $\sigma = \sqrt{d}\,m$ for $m \in \{0.25, 0.5, 1, 2, 4\}$, so for $d = 10$ latent dimensions the bandwidths are $\sigma = 0.79$, 1.58, 3.16, 6.32 and 12.65; they are fixed in advance and not tuned, and the MMD results hold for this kernel); debiased Sinkhorn divergence (entropy-regularised optimal transport; not W1 at the smoothing used; Euclidean cost, entropic smoothing $0.05\sqrt{2d}$ for latent dimension d, float64, marginal error 10^{-4}, cap of 500,000 iterations); scvi-tools adversarial classifier (conditioned decoder only; 32 hidden units, weight one minus the KL weight); plain scVI as strength zero; each critic head has two hidden layers of 128 units with one output per batch.

- Supplementary Note 2, training settings: two differences between methods are by design, the update steps and the optimiser (both below; the discriminator uses the generator's optimiser settings); plain scVI is the same model at strength zero; zero-inflated negative binomial likelihood; training split (90%); minibatch (128); optimisers (critic optimiser: Adam, learning rate 10^{-4}, betas 0 and 0.9; generator and discriminator: Adam, learning rate 10^{-3}, epsilon 0.01, weight decay 10^{-6}); update steps (one discriminator step, five critic steps per encoder step, by design; adversary steps reuse one minibatch); adversary input (posterior mean, no per-dimension standardisation) and the two input pilots, whose switching criteria were fixed before they were run (two tasks, both decoders, three seeds: a posterior sample as input lowered biology at matched batch removal by mean -0.004, 9 of 12 task–decoder–method combinations evaluable, three adversarial methods; standardisation by -0.003, 14 of 20 evaluable, five methods; neither met the criteria for switching); scvi-tools epoch rule ($\min(400, \mathrm{round}(20000 \times 400/n))$ epochs for n cells, no early stopping) with epochs, minibatch and latent size equal within task; epochs and final KL weight per task; KL warm-up of 400 epochs, incomplete on four shorter tasks, so the penalty acts from step one; 10 latent dimensions, 128 hidden units; all scoring on the posterior mean; failed fit defined as non-finite loss or latent. Values in Table S2.
- Supplementary Note 3, scIB metrics and scores: formulas of the five batch metrics (principal-component regression, batch average silhouette width, graph integration local inverse Simpson's index, graph connectivity, k-nearest-neighbour batch-effect test) and of each task's biology metrics, with the biology metric list per task; each fit is scored on its posterior-mean embedding; all metrics unscaled, raw 0–1, computed on all cells including training cells (the scIB convention; Li et al. instead report held-out cells; main-text reference); batch score, biology score, selection score (0.4 batch +0.6 biology, the scIB overall weights); local (three neighbourhood metrics) and global (two whole-embedding metrics) split with batch score $(3\,\mathrm{local} + 2\,\mathrm{global})/5$; input preparation: 2,000 highly variable genes, counts as distributed (non-integer in 18 batches of four tasks), chromatin-accessibility tasks as gene-activity matrices, one batch variable per task.
- Supplementary Note 4, penalty-strength sweep: design (plain scVI and the six penalties at the ten half-decade strengths 0.1, 0.3, 1, 3, 10, 30, 100, 300, 1,000 and 3,000 on all eight tasks, both decoders, seeds 100, 101 and 102, plus two edge strengths in two places: the discriminator with the unconditioned decoder also at 0.01 and 0.03, and the Sinkhorn divergence with the conditioned decoder also at 10,000 and 30,000; 378 fits per task (plain scVI 6; six penalties times ten strengths times three seeds times two decoders, 360; 12 edge-strength fits), 3,024 in all; plus nine fits at strength 0.03 for the discriminator, the reference critic and the pooled critic on the immune task with the conditioned decoder, from an earlier strength-extension round, so the analysis reads 3,033); why the edge strengths exist (a strength-choice rule for the planned fixed-strength benchmark reached the grid edge on the first three tasks fitted, the grid was extended by two half-decade values there, and the extension was kept for all eight tasks); the planned fixed-strength benchmark that the rules were to set up and that was not run, so the strength-choice rules choose no strength and the analysis reads every sweep strength (the replication's strengths, chosen from the sweep's batch scores, are described below); every fit of a task on one machine, so that hardware never differs between penalties within a task (the small chromatin-accessibility, immune and simulation 1 tasks on one local GPU, the other five tasks on the cluster; Supplementary Note 11); all Sinkhorn fits of the eight tasks were made with the corrected solver of 13 October 2026 (commit 10a1ca5): the 180 fits of the three local tasks were refitted on 17–19 October 2026 and the five cluster tasks were fitted from 22 October 2026. Replication across training seeds: design (for each of the 13 ranked task and decoder pairs and each of the six penalties, the two sweep strengths that bracket the first crossing of b^* on the sweep's seed-mean curve, refitted with five seeds (0–4); in one cell, the discriminator with the conditioned decoder on the human–mouse immune task, the bracket runs from plain scVI to the lowest sweep strength; plain scVI at seeds 0–4 on all eight tasks and both decoders; 855 fits, 775 with a penalty and 80 of plain scVI, and 40 fits of the extension step for 8 readings, 895 in all; the three low-signal pairs, immune, simulation 1 and simulation 2 with the conditioned decoder, skipped); 46 of the 78 sweep curves cross b^* more than once, and the first crossing sets the bracket; the extra outward strength and the bottom-rank rule when the new means do not bracket b^*; provenance (design script `experiments/sweep_replication/replication_design.py` with its run list and summary in `experiments/sweep_replication/design_20261102/`, commit 2113ec7; prediction file and

agreement rule at tag `replication-prediction-v1`, commit 3a90d0f, which record the sweep ranking table they were read from, analysis commit a361ba2, and its SHA-256).

- Supplementary Note 5, matched batch removal and the low-signal rule: why equal strength is not equal correction; b^* as the midpoint between plain scVI's mean batch score and the highest mean batch score all six penalties reach at strengths with no failed fit, per task and decoder (the highest common score and the curves use the same sweep strengths, so b^* lies below the highest common score, which every curve reaches, so with no failed fit every curve reaches b^* by construction; in the sweep 0 of the 96 task, decoder and penalty combinations lacked a reading, and the bottom-rank rule was never used); curve construction (plain scVI plus the sweep strengths, seed means over three seeds) and the interpolated reading where a curve first reaches b^*; leave-one-seed-out standard error of the reading at fixed b^* (the same seed left out of plain scVI) and range of the rank over the three repeats, reported only where at least two of the three readings exist and otherwise as not available; which pairs are ranked (every pair that is not low-signal; a penalty whose curve does not reach b^* takes the bottom rank); the low-signal rule (gap under 0.02, so b^* under 0.01 above plain scVI and not reliably placed) with the list of analyses that exclude low-signal pairs (ranking at b^*, hypervolume, local versus global reading) and keep them (share of non-dominated strengths, collapse rate); the separate table of low-signal readings; the immune task, plain scVI 0.638, highest common 0.657, gap 0.019; simulation 1, 0.812, 0.819, gap 0.007; simulation 2, 0.656, 0.665, gap 0.009; all with the conditioned decoder, none with the unconditioned decoder; matched strength (batch score nearest b^*) as used by label transfer; failed-fit counting.
- Supplementary Note 6, benchmark ranking: how biology at b^* is read for each penalty (curve of seed means over three seeds through plain scVI and the sweep strengths, first crossing, linear interpolation), its leave-one-seed-out standard error (formula; b^* held fixed, the same seed left out of plain scVI) and the range of its rank over the three repeats, reported only where at least two of the three leave-one-seed-out readings exist and otherwise as not available, the penalty being ranked either way; which pairs are ranked (every pair that is not low-signal); ranking rules (rank 1 is the highest biology at b^* among the six penalties, ranks within each task and decoder; tied penalties get the average of their ranks, so two tied for 2nd both get 2.5, also for penalties sharing the bottom rank; a penalty whose curve does not reach b^* takes the bottom rank in that pair and penalties that all fail share the last places, following scIB, where methods that could not be run on a task were given the same rank as unintegrated data (main-text reference); mean rank over all real tasks that are not low-signal per decoder as in scIB, decoders never combined); simulations ranked and reported separately; low-signal pairs excluded from the ranking and shown in their own table; related tasks (human–mouse immune contains all immune cells; large chromatin-accessibility contains the small task) counted as separate tasks; sensitivity analyses of the reading (per-run reading, biology without trajectory conservation on the two immune tasks, points with a failed fit removed) and rank stability under the scaling checks (Supplementary Figures S2 and S5); no significance tests, probabilities, margins or multiplicity corrections.
- Supplementary Note 7, rules for secondary analyses: hypervolume (reference point at plain scVI's batch score and 0.10 below its biology, division by the batch gap, low-signal excluded, reported per penalty, task and decoder); share of non-dominated strengths (dominance within the tie tolerance, a strength whose fits all failed is not a point, every task and decoder pair including the low-signal ones; Li et al.'s number of distinct choices is not used because it needs an arbitrary grid width); collapse (definition, 30 fits per penalty, task and decoder, 33 for the discriminator, the reference critic and the pooled critic on the immune task with the conditioned decoder (additional strengths), 36 where edge strengths were fitted, exact binomial 95% intervals, nominal because the fits of one seed are not independent, low-signal kept, failure types reported separately; an illustrative simulation of the interval's coverage (script `paper/analysis_pass10/coverage_sim_sweep.py`, random seed 7, 20,000 replicates per scenario). The estimand is the probability that the exact 95% interval computed from 30 fits (ten strengths, three seeds) or 36 fits (twelve strengths) contains the mean collapse probability over the strength grid. In scenario A the fits are independent and the collapse probability rises with strength along a profile set by hand (0.0, 0.0, 0.05, 0.2, 0.6, 0.95 at six points, interpolated linearly to ten or twelve strengths). In scenario B each seed has one collapse threshold shared by its strengths, so a seed that collapses at one strength collapses at every higher one; the threshold position is drawn from a distribution set by hand (0.0, 0.0, 0.0, 0.1, 0.15, 0.2, 0.55 over seven positions, the last meaning no collapse, interpolated and renormalised). A control with independent fits at a constant probability of 0.3 must cover at least 0.95. Coverage was at least 0.999 in scenario A, 0.88 (30 fits) and 0.83 (36 fits) in scenario B, and 0.97 and 0.95 in the control, with Monte Carlo standard errors of at most 0.003. Both profiles are illustrative choices, not estimated from the sweep; other dependence structures give other coverage, and the 33-fit design is not simulated, so the simulation shows that dependence can lower coverage below the nominal level but does not bound the coverage of the reported intervals); local versus global (metric split, reading at the same crossing as the biology reading, per penalty, descriptive); good-strength width (selection score, best strength, window at the lower end of the best strength's 95% confidence interval, width in decades, descriptive); single-strength placement (strength chosen on real tasks of other families, related tasks never choose for each other, only strengths with no failed fit in any choosing task eligible, ties to the smaller strength, simulations take the strength chosen on all six real

tasks, own best as optimistic bound, descriptive; the same rule picks the setting of Harmony, Scanorama and sysVI, Supplementary Note 8); best-of-k (expected best selection score among k random settings, $k = 1$–6, the number of settings of Harmony, Scanorama and sysVI; the per-task tables continue the penalties' curves to their ten or twelve strengths, but comparisons at equal tries stop at six, optimistic, scANVI in its own stratum, descriptive; computed per task, with Figure 6a showing the mean over the six real tasks of each method's curve and the per-task curves in Supplementary Figure S6; plain scVI, scANVI and the scvi-tools adversarial classifier have one setting each, so plain scVI is the $k = 1$ reference for each penalty's selection score).

- Supplementary Note 8, baselines: Harmony, Scanorama and sysVI at six settings each (Harmony's diversity penalty from 0 to 8 at 10 principal components, Scanorama's neighbour count from 2 to 80 at 10 dimensions, sysVI's cycle-consistency weight from 1 to 50); scANVI (uses cell-type labels) and principal-component analysis (uncorrected reference) at one setting, so neither involves a choice; in the single-strength placement each of the three tools gets its setting by the same rule as the penalties (the setting with the best mean selection score on real tasks of other families, related tasks never choosing for each other, with the task's own best setting in parentheses as the optimistic bound); five seeds (0–4) for the neural methods, one run otherwise; 448 fits on the eight tasks (scvi-tools adversary 40, scANVI 40, sysVI 240, Harmony 56, Scanorama 56, PCA 16) including three sensitivity settings (Harmony at 50 principal components, Scanorama at 100 dimensions, PCA at 50 components); grids and settings (Table S4); baseline grid amendments.
- Supplementary Note 9, label transfer: matched-strength embeddings for three seeds (the strength whose batch score is nearest b^*, never chosen by accuracy); plain scVI and the scvi-tools adversarial classifier unmatched; each batch held out in turn; 15-nearest-neighbour classifier; cell types in one batch only dropped, from labels alone; F1 per cell type averaged over types; gap to b^* beside each value; low-signal pairs tabulated but not summarised; no test; the full label-transfer table.
- Supplementary Note 10, rules fixed before scoring and deviations: when the sweep rules and the ranking-analysis rules were fixed, and which sweep scores had been read at that time (sweep rules: amendments of 4 and 16 October 2026, the last at commit 10a1ca5, the five further tasks added by the pilot extension decided on 21 October with those rules; ranking rules: the descriptive ranking design written on 22 and 23 October 2026, commits 065a486 and 5d01da0; by then the strength-choice rules had read the seed means of the small chromatin-accessibility, immune and simulation 1 tasks, and the other five tasks were fitted from 22 October and scored from 28 October; the analysis code for all eight tasks was written on 1 November, with a first run on all eight tasks that day for debugging, and the hypervolume and the share of non-dominated strengths were added on 2 November, after those scores had been read); changes from the ranking specification (three seeds instead of five, every fitted strength a point, b^* computed from the sweep rows, at least two of three leave-one-seed-out readings); that the planned fixed-strength benchmark the rules were to set up was not run, so the strength-choice rules selected nothing; amendment list; early-scored fits (160 fits of the planned fixed-strength benchmark were scored, 19 on 12 October and 141 on 20–21 October 2026, on the immune task with the batch-conditioned decoder, and are not used; about 572 fits were made before the benchmark was stopped on 21 October: 167 on the workstation and 405 on the cluster, the cluster fits not scored); grid-edge extension fits; post-pilot design choices (adversary input, standardisation); baseline grid amendments; seed pairing; the ranking analysis run once, after all sweep and baseline fits are scored or failed; deviations from the plan fixed in advance; the long limitations list. For the replication across training seeds: the sweep readings and ranks at b^* for the 13 ranked pairs were written to a prediction file and the agreement rule fixed, both committed under one tag, before any replication fit was scored (tag and commit in Supplementary Note 4); the raw new-seed reading is primary and a slope-adjusted reading is a sensitivity analysis only; b^* and the replication strengths were set from the sweep's batch scores after the sweep had been read.
- Supplementary Note 11, software, hardware, run counts and checksums: scvi-tools and package versions; hardware; run counts per experiment (strength sweep 3,024 including 96 edge-strength fits (12 per task), plus nine strength-extension fits on the immune task (3,033 analysed); planned fixed-strength benchmark not run, about 572 fits made for it before it was stopped on 21 October 2026 (167 on the workstation, 405 on the cluster; the cluster count is the repository record's, not re-counted), unused; baselines 448; input and standardisation pilots) with completed and failed fits (Table S5); a fit fails if its training loss or its latent becomes non-finite; checksums of task files and results; repository and version tag. Scaling checks (Supplementary Figure S5). Replication across training seeds: 855 fits (775 with a penalty, 80 of plain scVI), plus 40 fits of the extension step (15 on the workstation, 25 on the cluster), 895 in all, 0 failed, completed and failed counts in Table S5; each task fitted and scored on the same kind of machine as its sweep fits (the small chromatin-accessibility, immune and simulation 1 tasks on the workstation GPU, the other five tasks on the cluster's L40S GPUs), with the same scorer and a fixed seed for the local inverse Simpson's index.
- Supplementary Note 12, direct biology check: exploratory and descriptive; the rules were written after the sweep scores (batch and biology scores, b^*, matched strengths, label-transfer macro F1) had been read and before any value of this check was computed; no new fits, no test. Fits: for each penalty, decoder and task, the label-transfer strength (the failure-free strength whose seed-mean batch score is nearest b^*, ties to the smaller strength), seeds

100–102; plain scVI with both decoders, seeds 100–102; every scored baseline row at each of its settings (783 fits on eight tasks; low-signal pairs included and flagged). Neighbours: the 15 nearest other cells by Euclidean distance in the saved latent. Readouts per fit and cell type: neighbourhood purity (share of the 15 neighbours with the cell's own label, averaged over the type's cells; per fit, the unweighted mean over types); rare types, under 1% of a task's cells, with their purity and label-transfer recall; marker-gene preservation on the four RNA tasks (log-normalised expression as distributed, restricted to the 2,000 highly variable genes; top 20 Wilcoxon markers per type against all other cells, computed once on the expression as distributed and once on each fit's expression averaged over the 15 neighbours; Jaccard overlap of the two sets; types under 10 cells get no marker readout); label-transfer confusion from the classifier of Supplementary Note 9 (each batch held out in turn, 15-nearest-neighbour classifier, types present in one batch only dropped), with per-type recall and the row-normalised confusion table. Differences: penalties and plain scVI against plain scVI with the same decoder and seed; baselines against the mean of plain scVI with the conditioned decoder over seeds 100–102; a fit's difference is the mean of its per-type differences. Summaries: per task, method, setting and decoder, the mean and SD over seeds, then the mean over the real tasks (Table S10). scANVI is trained with the cell-type labels, so its readouts are not label-free. Checks that stop the run: latent checksums, cell order and labels against the task file, fit counts, seed-matched references, readouts within [0, 1], and the macro F1 recomputed from the confusion predictions equal to the label-transfer F1 of Supplementary Note 9. Code commit c4a793f, branch `biology-check`.

Supplementary Tables

- Table S1, tasks: for each of the eight tasks, cells, batches, cell types, genes (2,000 highly variable), real or simulated, family and nesting, count type as distributed (non-integer flag), gene-activity for chromatin-accessibility tasks, checksums.
- Table S2, hyperparameters and training settings: latent dimensions (10), hidden units (128), likelihood (zero-inflated negative binomial), training split (90%), minibatch, epochs per task from the scvi-tools rule, final KL weight per task, warm-up (400 epochs), optimisers, update steps, gradient penalty weight (10), adversary input, MMD kernel scales, Sinkhorn smoothing, seeds per experiment.
- Table S4, baseline grids: the six settings of Harmony, Scanorama and sysVI, the single settings of scANVI and principal-component analysis, seeds, sensitivity settings.
- Table S5, run counts: fits planned, completed and failed per experiment, method and decoder.
- Table S6, ranking detail: biology at b^* per penalty, task and decoder with leave-one-seed-out standard errors, ranks and the range of each rank over the three repeats, a separate table of low-signal pairs, and the pairs where a penalty's curve does not reach b^* marked.
- Table S7, single-strength placement for the unconditioned decoder (the counterpart of Table 3): selection score per method and real task at the strength (for Harmony, Scanorama and sysVI, the setting) chosen on other task families, with the task's own best strength or setting in parentheses.
- Table S8, secondary results per penalty, task and decoder: hypervolume, share of non-dominated strengths, collapse rate with intervals and failure types, good-strength width and number of good strengths, local and global readings at b^*.
- Table S9, replication across training seeds: per penalty, task and decoder, the two bracketing strengths, the batch and biology means over seeds 0–4 at each, the batch score reached, biology at b^* on the new seeds beside the sweep reading, the replication rank beside the sweep rank, and cells that needed the extra outward strength or did not replicate the crossing; Kendall's τ_b per pair; the four large sweep gaps with their sign on the new seeds; plain scVI at seeds 0–4 beside the sweep's plain scVI per task and decoder.
- Table S10, direct biology check: per method and decoder, the mean over the six real tasks of neighbourhood purity minus plain scVI's with its seed spread, rare-type purity and label-transfer recall minus plain scVI's, and marker-gene Jaccard overlap with its difference from plain scVI (typeset after Table 3); per-task, per-cell-type and confusion tables in the repository.

Supplementary Figures

- Figure S1, simulation tasks: trade-off curves and per-task ranking at b^*, each simulation on its own.
- Figure S2, sensitivity: ranking under alternative readings (per-run reading, biology without trajectory conservation on the two immune tasks, points with a failed fit removed).
- Figure S3, label transfer: F1 per cell type, method and task at matched strength, with the gap to b^*.
- Figure S4, per-metric strength curves: each batch and biology metric against strength, per method and decoder.
- Figure S5, scaling checks.
- Figure S6, best of k settings per task: each method's expected best selection score among k settings drawn at random, $k = 1$ to 6, one panel per real task; the mean of these curves over the six real tasks is Figure 6a.

Figures and tables

400 words of captions and notes

- Figure 1: (a) model schematic: scVI encoder and decoder with or without the batch label, swappable alignment term; (b) how biology is read at matched batch removal b^*. Second cut, after Figure 5.
- Figure 2: ranking at matched batch removal: (a) heatmap of biology score at b^* (colour) and rank (number) for the six penalties in each real task, conditioned and unconditioned decoder, not-ranked pairs hatched; (b) mean rank over the real tasks that are not low-signal per penalty and decoder with the range across tasks.
- Figure 3: biology score against batch score trade-off curves, decoders by real tasks, line per method, plain scVI point, b^* line, standard tools as points.
- Figure 4: per-penalty dot plots, one point per real task and the mean: (a) hypervolume; (b) collapse rate with 95% intervals; (c) local minus global batch removal at b^*.
- Figure 5: good-strength width per penalty and decoder, one point per task. First to move to the supplement if pages run short.
- Figure 6: (a) expected best selection score against number of tries, mean over the six real tasks of each method's per-task curve; (b) batch against biology score at single strength.
- Table 1: methods: alignment term, network, update steps, sweep strengths.
- Table 2: benchmark ranking: mean rank per penalty and decoder over the real tasks that are not low-signal, range over tasks, mean biology at b^*, simulation ranks, tasks not ranked, tasks ranked.
- Table 3: single-strength placement: selection score per method, decoder, task.

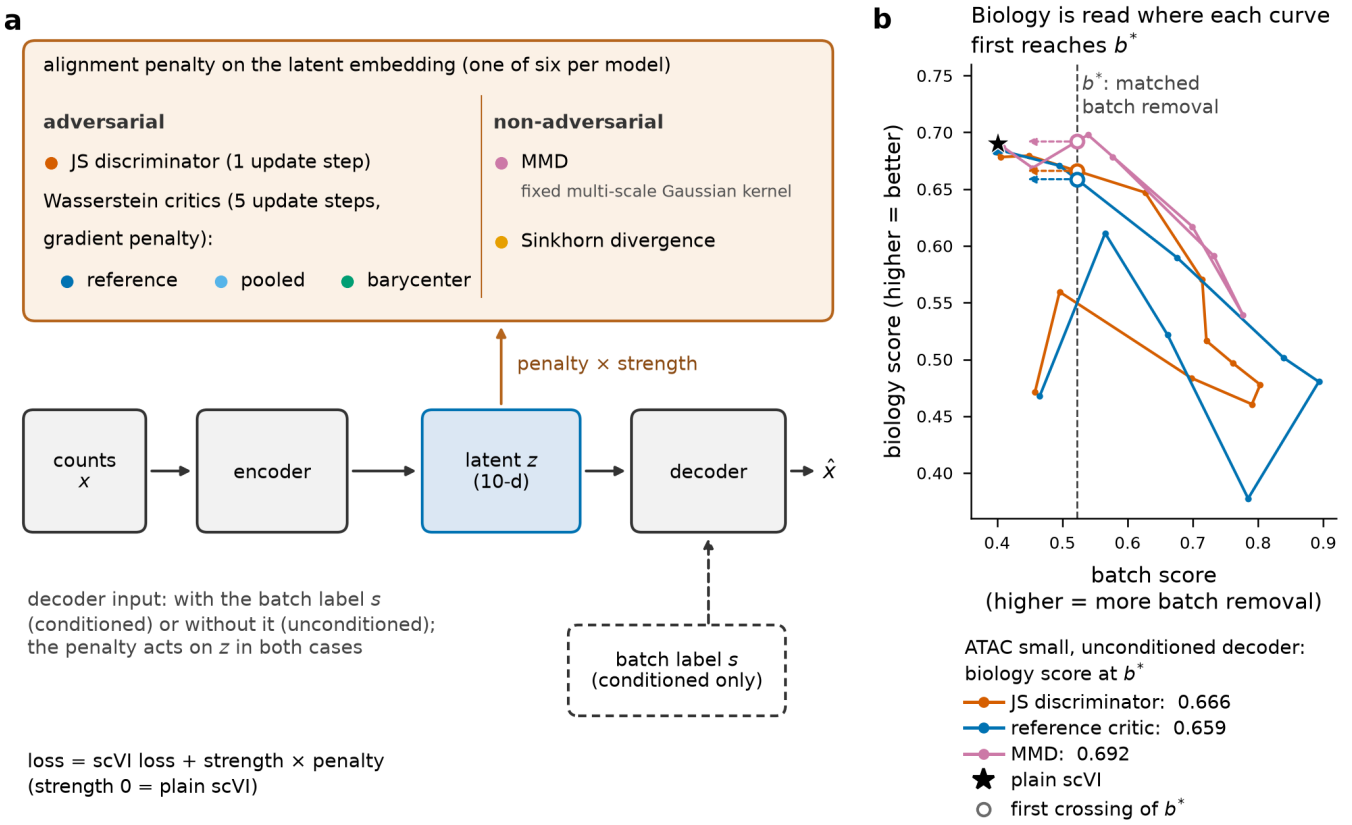


Figure 1: Model and read-out. (a) One scVI encoder–decoder is shared by all methods; the decoder receives the batch label (conditioned) or not (unconditioned), and only the alignment penalty on the 10-dimensional latent embedding is swapped. Four adversarial penalties (a Jensen–Shannon discriminator, one update step; Wasserstein critics, reference, pooled and barycenter, five update steps with a gradient penalty) and two non-adversarial penalties (MMD with a fixed multi-scale Gaussian kernel; the Sinkhorn divergence) are compared. The loss is the scVI loss plus the strength times the penalty; strength 0 is plain scVI. (b) One real task and decoder from the sweep, ATAC small with the unconditioned decoder, for three of the six penalties (colours as in Figures 2–6). Each curve joins plain scVI (star) and the sweep strengths in increasing order (10 strengths for the reference critic and MMD, 12 for the discriminator, which has two further low strengths); each point is a mean over the three seeds. Biology is read where the curve first reaches the matched batch removal b^* (dashed line, 0.522), interpolating between neighbouring strengths (open circles; dotted arrows read the biology score): JS discriminator 0.666, reference critic 0.659, MMD 0.692. The discriminator and reference-critic curves cross b^* a second time at higher strengths; the reading uses the first crossing.

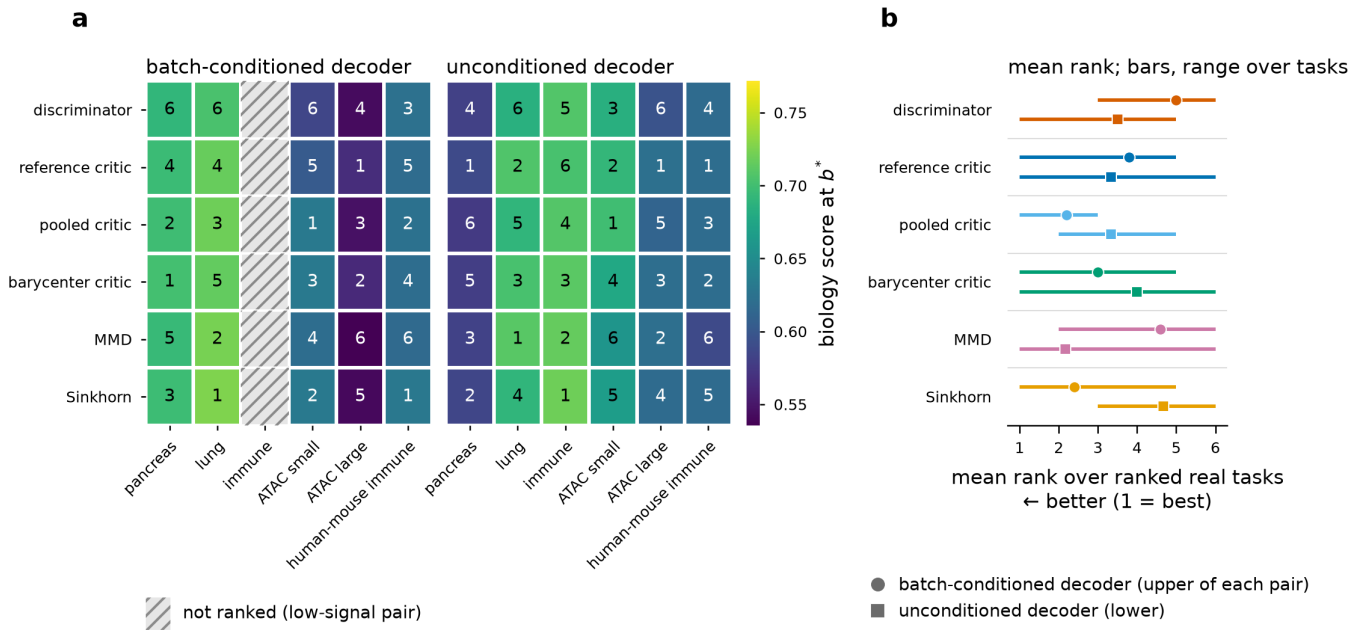


Figure 2: Ranking of the six penalties by biology kept at matched batch removal b^*. (a) Biology score at b^* (colour) and rank among the six penalties within each task column (number; tied penalties share the average rank) for each real task, with the batch-conditioned and the unconditioned decoder. Hatched cells: the immune task with the batch-conditioned decoder, a low-signal pair that is not ranked. A penalty whose curve does not reach b^* would take the bottom rank; every curve reached it here. (b) Mean rank over the ranked real tasks (five with the batch-conditioned decoder, six with the unconditioned decoder) per penalty and decoder; rank 1 is best; bars span the lowest and highest rank over those tasks. The pooled critic had the best mean rank with the batch-conditioned decoder (2.2) and MMD with the unconditioned decoder (2.17). The leave-one-seed-out standard error of each reading in (a) and the range of each rank over the three leave-one-seed-out repeats are in Supplementary Table S6. Simulations are ranked separately (Supplementary Figure S1).

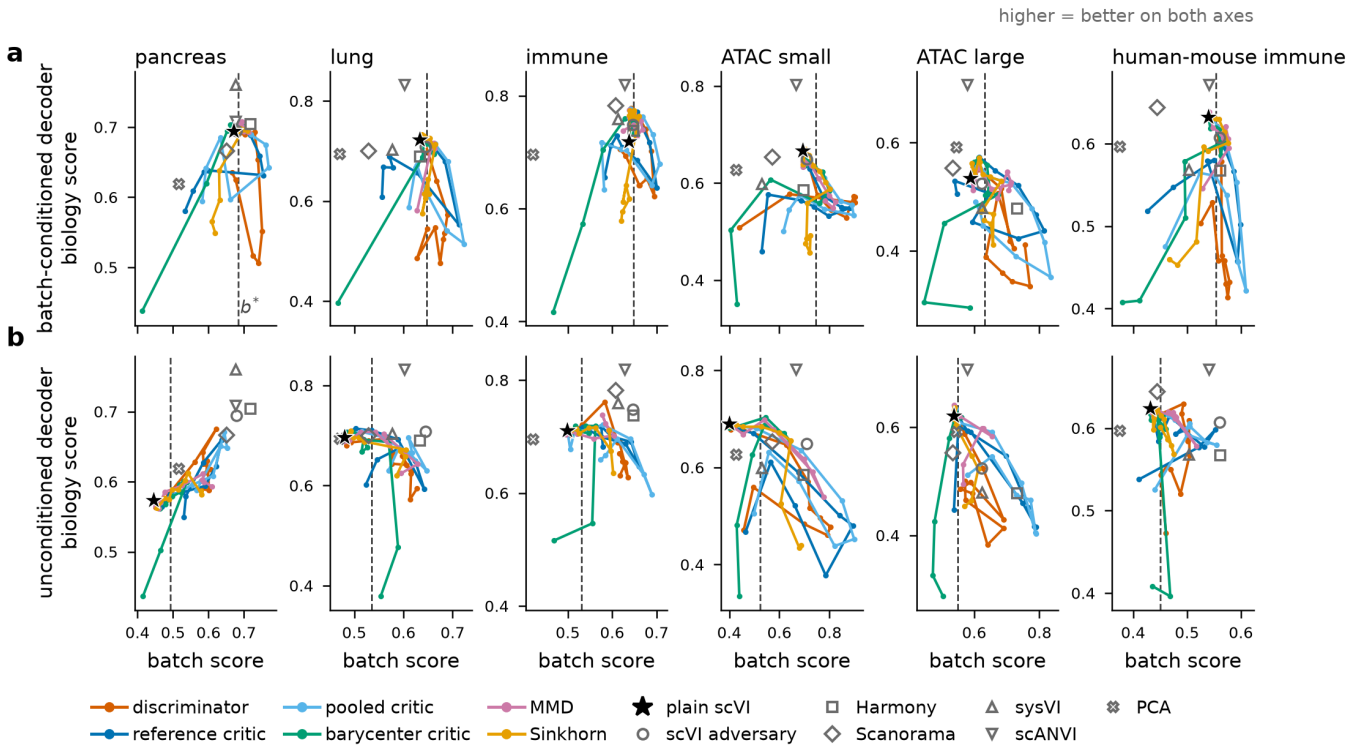


Figure 3: Batch score against biology score for the six real tasks (columns) with the batch-conditioned decoder (a) and the unconditioned decoder (b). Each line joins plain scVI (star) and one penalty's fitted strengths in increasing order, by mean scores over three seeds; no real-task curve ended early because of failed fits. Dashed line, matched batch removal b^*. Grey open symbols, each standard tool at its best setting by selection score (0.4 batch plus 0.6 biology) among the settings run; scANVI uses cell-type labels. Panels in a column share axes.

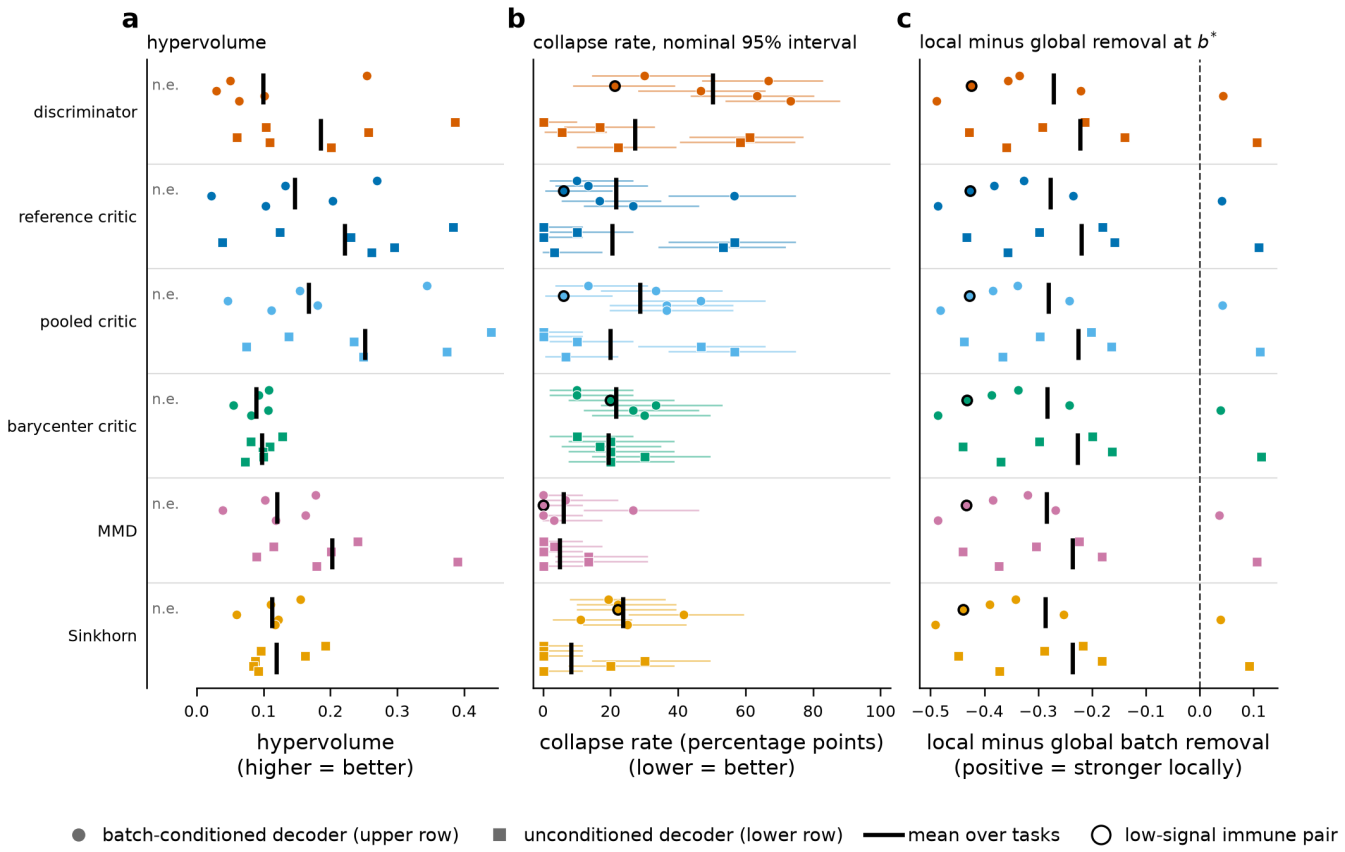


Figure 4: Secondary summaries per penalty with the batch-conditioned (circles, upper row of each band) and unconditioned decoder (squares, lower row). One point per real task, in the order of Figure 3 from top to bottom within each row; black tick, mean over tasks. (a) Hypervolume of the strength curve (higher is better); it is not evaluable (n.e.) for the low-signal immune pair with the batch-conditioned decoder, so that point is absent and the mean is over five tasks. (b) Collapse rate in percentage points with nominal 95% intervals (lower is better); the mean includes the low-signal pair. (c) Local minus global batch removal at b^*; positive means stronger local removal; the mean excludes the low-signal pair. Ringed point, the low-signal immune pair with the batch-conditioned decoder.

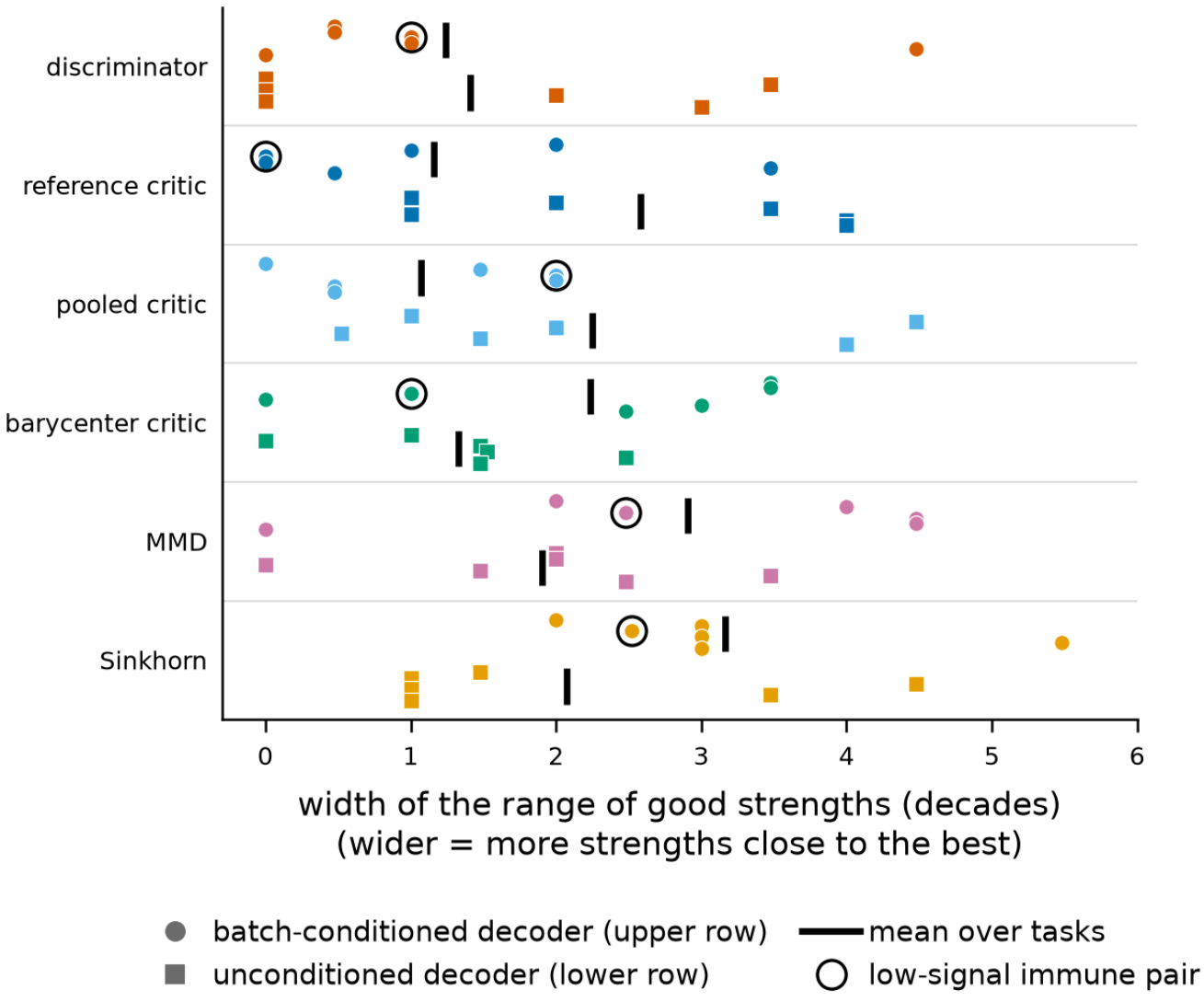


Figure 5: Width of each penalty's range of good strengths, in decades, one point per real task and decoder (batch-conditioned decoder, circles, upper row of each band; unconditioned decoder, squares, lower row; tasks in the order of Figure 3 from top to bottom; black tick, mean over tasks). A good strength has a mean selection score at least the lower 95% confidence bound of the best strength; a width of zero means a single qualifying strength. Ringed point, the low-signal immune pair with the batch-conditioned decoder, included as in the Results text.

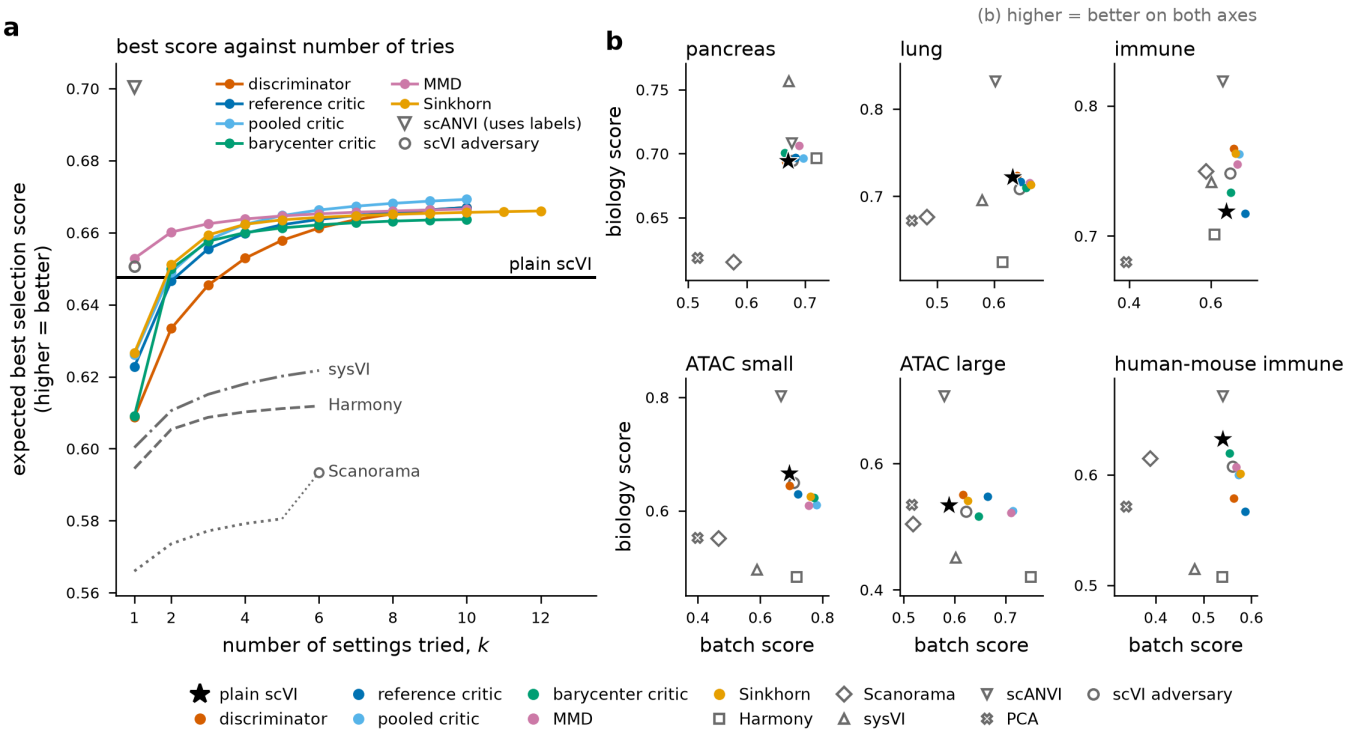


Figure 6: Placement against plain scVI and standard tools with the batch-conditioned decoder. (a) Expected best selection score among k settings drawn at random, mean over the six real tasks of each method's per-task curve (Supplementary Figure S6; selection score: 0.4 batch plus 0.6 biology; optimistic, because the best setting is chosen and scored on the same data). Each curve is drawn up to the largest k for which all six tasks have k settings: $k = 10$ for five penalties (the discriminator, reference critic and pooled critic have an eleventh setting on the immune task only) and $k = 12$ for the Sinkhorn divergence, and $k = 6$ for Harmony and sysVI. Scanorama has six settings on five tasks and five on the large chromatin-accessibility task (80 neighbours could not be fitted); it is drawn to $k = 5$, and its $k = 6$ value (open marker) is the mean of five tasks. The Results compare methods at $k = 1$ and $k = 6$, the number of settings of Harmony, Scanorama and sysVI; the penalty curves beyond $k = 6$ are shown for completeness. Horizontal line, plain scVI (0.648); scANVI (uses cell-type labels) and the scVI adversary have one setting. At $k = 1$ every penalty except MMD scored below plain scVI; the other penalties first exceeded plain scVI at $k = 2$ (pooled critic, barycenter critic, Sinkhorn divergence), 3 (reference critic) and 4 (discriminator). (b) Batch against biology score for each real task at the single strength (penalties, Harmony, Scanorama, sysVI) chosen on other task families, or at the fixed setting (scANVI, scVI adversary, PCA with 10 dimensions); star, plain scVI. The unconditioned decoder is in Supplementary Table S7.

Table 1: Alignment methods compared on the shared scVI backbone. Each penalty is multiplied by a fixed strength and added to the scVI loss. W1, Wasserstein-1; JS, Jensen–Shannon; MMD, maximum mean discrepancy; OT, optimal transport; KL, Kullback–Leibler. Equations in Supplementary Note 1; training settings in Supplementary Note 2 and Table S2.

Method	Alignment term	Network	Adversary steps per encoder step	Gradient penalty or regularisation	Sweep strengths (Methods)
JS discriminator	scvi-tools fool loss (JS-type): encoder trained to make the batch classifier wrong	batch classifier, two hidden layers of 128 units, one output per batch	1	none (plain)	0.1–3,000; also 0.01 and 0.03 with the unconditioned decoder
Reference critic	W1 distance of each batch to one reference batch (highest cell-type entropy)	critic, two hidden layers of 128 units, one output per batch	5	gradient penalty, weight 10	0.1–3,000
Pooled critic	W1 distance of each batch to all other batches pooled	critic, as above	5	gradient penalty, weight 10	0.1–3,000
Barycenter critic	W1 distance of each batch to the minibatch Wasserstein barycenter (an average distribution)	critic, as above	5	gradient penalty, weight 10	0.1–3,000
MMD	kernel two-sample distance between batches, multi-scale Gaussian kernel	none	–	none	0.1–3,000
Sinkhorn	debiased Sinkhorn divergence (entropy-regularised OT); not W1 at this smoothing	none	–	entropic smoothing (Supplementary Table S2)	0.1–3,000; also 10,000 and 30,000 with the conditioned decoder
scvi-tools adversary	scvi-tools adversarial batch classifier (reference method, conditioned decoder only)	batch classifier (scvi-tools default, two hidden layers of 32 units)	1	none; its weight is one minus the KL weight	no strength (weight fixed)

Table 2: Benchmark ranking of the six penalties by biology score at matched batch removal b^*. Mean rank over the real tasks that are not low-signal (rank 1 is best; tied penalties share the average rank, and a penalty whose curve does not reach b^* takes the bottom rank) and its range over tasks; mean biology score at b^*; ranks in simulation 1 and simulation 2, which are ranked separately; real tasks not ranked (low-signal) and number of real tasks ranked, of six. Per-task values are in Supplementary Table S6. Range, lowest and highest rank over the ranked real tasks; biology, mean over those tasks of the biology score at b^*; the simulations are low-signal with the conditioned decoder (–); ranked, real tasks ranked of six.

Penalty	Decoder	Mean rank	Range	Biology	Sim. 1 rank	Sim. 2 rank	Not ranked	Ranked
JS discriminator	conditioned	5.00	3–6	0.629	–	–	immune	5/6
	unconditioned	3.50	1–5	0.649	1	1	none	6/6
Reference critic	conditioned	3.80	1–5	0.640	–	–	immune	5/6
	unconditioned	3.33	1–6	0.646	2	4	none	6/6
Pooled critic	conditioned	2.20	1–3	0.644	–	–	immune	5/6
	unconditioned	3.33	2–5	0.651	5	2	none	6/6
Barycenter critic	conditioned	3.00	1–5	0.646	–	–	immune	5/6
	unconditioned	4.00	1–6	0.650	6	5	none	6/6
MMD	conditioned	4.60	2–6	0.638	–	–	immune	5/6
	unconditioned	2.17	1–6	0.655	3	3	none	6/6
Sinkhorn	conditioned	2.40	1–5	0.645	–	–	immune	5/6
	unconditioned	4.67	3–6	0.645	4	6	none	6/6

Table 3: Single-strength placement: mean selection score over seeds (three for plain scVI and the penalties; five for the scvi-tools adversary, scANVI and sysVI; one run for Harmony, Scanorama and PCA) (0.4×batch + 0.6×biology) per method and real task, conditioned decoder, at the strength with the best mean selection score on real tasks of other families (related tasks never choose for each other). Values in parentheses use the task's own best strength (optimistic bound). Harmony, Scanorama and sysVI are placed by the same rule, at the setting chosen on real tasks of other families, with the task's own best setting in parentheses; scANVI and principal-component analysis (PCA) run at one setting, so no choice is made; scANVI uses cell-type labels. Unconditioned decoder in Supplementary Table S7. Scanorama on the large chromatin-accessibility task has five of its six settings (80 neighbours could not be fitted). The immune task is low-signal with this decoder. Values are rounded to three decimals: the Sinkhorn divergence on pancreas (0.6846) is 0.0004 below plain scVI (0.6850) and counts as not exceeding it.

Method	Pancreas	Lung	Immune	ATAC small	ATAC large	Human–mouse immune
Plain scVI	0.685	0.686	0.686	0.677	0.556	0.595
scvi-tools adversary	0.688	0.683	0.708	0.673	0.564	0.588
JS discriminator	0.683 (0.708)	0.690 (0.690)	0.723 (0.724)	0.664 (0.703)	0.577 (0.582)	0.573 (0.595)
Reference critic	0.692 (0.700)	0.689 (0.690)	0.704 (0.725)	0.666 (0.682)	0.595 (0.606)	0.575 (0.598)
Pooled critic	0.697 (0.709)	0.694 (0.694)	0.725 (0.725)	0.679 (0.689)	0.601 (0.601)	0.589 (0.597)
Barycenter critic	0.687 (0.700)	0.689 (0.693)	0.700 (0.721)	0.683 (0.683)	0.569 (0.591)	0.593 (0.594)
MMD	0.700 (0.702)	0.694 (0.694)	0.719 (0.721)	0.668 (0.672)	0.598 (0.608)	0.591 (0.601)
Sinkhorn	0.685 (0.701)	0.694 (0.697)	0.722 (0.724)	0.679 (0.683)	0.575 (0.591)	0.591 (0.602)
Harmony	0.706 (0.710)	0.621 (0.621)	0.664 (0.664)	0.576 (0.587)	0.552 (0.558)	0.520 (0.531)
Scanorama	0.600 (0.613)	0.598 (0.600)	0.685 (0.692)	0.517 (0.538)	0.510 (0.522)	0.524 (0.525)
sysVI	0.723 (0.726)	0.649 (0.653)	0.685 (0.700)	0.533 (0.571)	0.511 (0.537)	0.501 (0.542)
scANVI (labels)	0.696	0.740	0.743	0.748	0.656	0.619
PCA (uncorrected)	0.577	0.585	0.565	0.490	0.527	0.478

Supplementary Table S10: Direct biology check (exploratory; rules written after the sweep scores had been read; Supplementary Note 12). Means over the six real tasks of per-task seed means. Purity and rare-type columns are differences from plain scVI (penalties and plain scVI: same decoder and seed; baselines: conditioned plain scVI, mean of seeds 100–102). Seed SD, mean over the six real tasks of the SD of the purity difference over seeds (n/a for one-run tools). Rare types on the four real tasks that have them; marker-gene Jaccard on the four RNA tasks. Baselines with several settings: mean over settings (number of settings in parentheses; Scanorama has six on the large chromatin-accessibility task). scANVI uses cell-type labels.

Method	Decoder	Purity diff.	Seed SD	Rare purity diff.	Rare recall diff.	Marker Jaccard	Jaccard diff.
JS discriminator	conditioned	−0.052	0.030	−0.107	−0.121	0.398	0.000
	unconditioned	−0.077	0.028	−0.076	−0.129	0.392	+0.001
Reference critic	conditioned	−0.004	0.008	−0.002	+0.005	0.399	0.000
	unconditioned	−0.011	0.011	+0.002	+0.008	0.394	+0.003
Pooled critic	conditioned	−0.019	0.008	−0.024	−0.036	0.398	−0.001
	unconditioned	+0.001	0.005	+0.010	+0.011	0.390	−0.001
Barycenter critic	conditioned	−0.007	0.007	−0.004	+0.001	0.395	−0.003
	unconditioned	+0.001	0.007	+0.010	+0.005	0.391	0.000
MMD	conditioned	−0.001	0.006	+0.003	+0.007	0.397	−0.001
	unconditioned	−0.004	0.005	+0.001	+0.003	0.394	+0.003
Sinkhorn	conditioned	−0.022	0.008	+0.006	+0.002	0.394	−0.005
	unconditioned	−0.033	0.008	−0.045	−0.079	0.397	+0.006
Plain scVI	conditioned	0.000	0.000	0.000	0.000	0.399	0.000
	unconditioned	0.000	0.000	0.000	0.000	0.391	0.000
scvi-tools adversary (1)	conditioned	−0.007	0.009	−0.001	+0.013	0.398	−0.001
scANVI (labels) (1)	–	+0.115	0.003	+0.069	+0.061	0.390	−0.008
sysVI (6)	–	−0.040	0.023	−0.045	−0.078	0.392	−0.007
Harmony (7)	–	−0.141	n/a	−0.234	−0.330	0.367	−0.031
Scanorama (7)	–	−0.049	n/a	−0.076	−0.087	0.382	−0.017
PCA (2)	–	−0.021	n/a	−0.062	−0.088	0.371	−0.028

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Supplementary content (Note 1, Tables S6, S7 and S9, Figures S1–S6)

Supplementary Note 1. Model and loss equations.

Notation. A cell i has counts x_{ig} for genes $g = 1, \dots, G$ (2,000 highly variable genes, or all features for the chromatin-accessibility tasks), a batch label $s_i \in \{1, \dots, K\}$ and a latent vector $z_i \in \mathbb{R}^d$ with $d = 10$. The encoder gives $q(z_i \mid x_i) = \mathcal{N}(\mu_i, \operatorname{diag} \sigma_i^2)$. $\mathcal{B}$ is a minibatch of 128 cells, $\mathcal{B}_k$ its cells from batch k and $\mathcal{B}_{-k}$ its cells from the other batches. $K_{\mathcal{B}}$ is the set of batches for which both $\mathcal{B}_k$ and $\mathcal{B}_{-k}$ are non-empty; a minibatch with no such batch adds zero to the penalty. Every penalty is applied to the posterior mean, $u_i = \mu_i$ (no per-dimension standardisation); $\hat{P}_k$ is the empirical distribution of $\{u_i : i \in \mathcal{B}_k\}$ and $\hat{P}_{-k}$ that of $\{u_i : i \in \mathcal{B}_{-k}\}$. The strength is $\lambda \geq 0$; plain scVI is $\lambda = 0$.

Encoder objective. For all methods the encoder and decoder minimise, over each minibatch,

$$\mathcal{L}_G \;=\; \mathcal{L}_{\text{scVI}} \;+\; \lambda\, \mathcal{A}(u_{\mathcal{B}}), \qquad \mathcal{L}_{\text{scVI}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \Big[- \mathbb{E}_q \sum_g \log p_{\text{ZINB}}(x_{ig} \mid \mu_{ig}, \theta_g, \pi_{ig}) + \beta_e \operatorname{KL}\big(q(z_i \mid x_i) \,\|\, \mathcal{N}(0, I_d)\big) \Big], \tag{1}$$

where $\mathcal{A}$ is the alignment term of the penalty (below), the expectation is over the posterior of the latent, and the likelihood is the zero-inflated negative binomial

$$p_{\text{ZINB}}(x \mid \mu, \theta, \pi) = \pi\, \mathbf{1}[x = 0] + (1 - \pi)\, \frac{\Gamma(x + \theta)}{\Gamma(\theta)\, x!} \Big(\frac{\theta}{\theta + \mu} \Big)^{\theta} \Big(\frac{\mu}{\theta + \mu} \Big)^{x}, \qquad \mu_{ig} = \ell_i\, \rho_g(z_i, s_i), \quad \ell_i = \textstyle\sum_{g'} x_{ig'}, \tag{2}$$

with ℓ_i the observed library size (scvi-tools' default), $\rho(z_i, s_i)$ the decoder's softmax over genes, θ_g a per-gene dispersion and π_{ig} the dropout probability produced by the decoder. The encoder reads $\log(1 + x_i)$. In the batch-conditioned setting the decoder receives the one-hot batch label s_i; in the unconditioned setting no batch label reaches the encoder or the decoder. The weight of the latent KL term follows scvi-tools' linear warm-up, $\beta_e = e/400$ for epochs $e < 400$ and 1 afterwards (the final value for each task is in Table S2). The penalty acts from the first step and is not scaled by the warm-up. The adversary heads below are trained on detached $u_{\mathcal{B}}$, and only the encoder gradient of $\mathcal{A}$ enters $\mathcal{L}_G$.

Jensen–Shannon-type discriminator. A classifier head $h : \mathbb{R}^d \to \mathbb{R}^K$ (two hidden layers of 128 units) is trained with one step per encoder step to predict the batch,

$$\min_h \; \mathcal{L}_D = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \mathcal{L}_{\text{CE}}\big(h(u_i), s_i\big), \qquad \mathcal{L}_{\text{CE}}(h(u), s) = -\log \operatorname{softmax}(h(u))_s, \tag{3}$$

and the encoder minimises the bounded fool loss of scvi-tools, the cross-entropy between the prediction and the uniform distribution over the other $K - 1$ batches,

$$\mathcal{A}_{\text{JS}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \frac{1}{K - 1} \sum_{k \neq s_i} \log \operatorname{softmax}\big(h(u_i)\big)_k. \tag{4}$$

Wasserstein-1 critics. A critic head $C = (C_1, \dots, C_K)$ (two hidden layers of 128 units, one output per batch) estimates, for each active batch k with target distribution T_k, the Kantorovich–Rubinstein dual

$$\hat{W}_k = \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} C_k(u_i) - \frac{1}{|T_k|} \sum_{t \in T_k} C_k(t) \;\approx\; W_1(\hat{P}_k, \hat{T}_k), \tag{5}$$

where the right-hand side is the distance between the empirical distributions; the dual is a lower bound of it for a 1-Lipschitz C_k, which the gradient penalty below enforces softly, so $\hat{W}_k$ is an estimate and not a bound. The targets are: reference critic, $T_k = \{u_j : s_j = r\}$, the cells of the reference batch r, the batch with the highest cell-type entropy $-\sum_c p_{r,c} \log p_{r,c}$ (head r is inactive; this rule uses the cell-type labels); pooled critic, $T_k = \{u_j : j \in \mathcal{B}_{-k}\}$; barycenter critic, $T_k = \hat{B}$, the support of the Wasserstein-2 barycenter of the batches present in the minibatch,

$$\hat{B} = \arg\min_B \; \frac{1}{K_{\text{pres}}} \sum_{k \text{ present}} W_2^2(\hat{P}_k, B), \tag{6}$$

with $|\mathcal{B}|$ atoms of equal weight, solved by the free-support fixed-point algorithm (at most 10 iterations, stopping threshold 10^{-7}), started from the minibatch cells and recomputed at every step; $\hat{B}$ is detached, so no gradient passes through it. A head is active when its batch is present in the minibatch and its target is non-empty; $\mathcal{H}$ is the set of active heads. The critic takes five steps per encoder step on

$$\min_C \; \mathcal{L}_C = -\frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \hat{W}_k \;+\; \lambda_{\text{GP}}\, \frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} \Big(\big\| \nabla_x C_k(\hat{x}_i) \big\|_2 - 1 \Big)^2, \qquad \hat{x}_i = \varepsilon_i u_i + (1 - \varepsilon_i) t_i, \tag{7}$$

with $\lambda_{\text{GP}} = 10$, $\varepsilon_i \sim \text{U}(0, 1)$ and t_i drawn uniformly from T_k for each cell i of batch k. In the encoder step the gradient penalty is dropped and the encoder minimises the critic's estimate,

$$\mathcal{A}_W = \frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \hat{W}_k, \tag{8}$$

where the reference cells (reference critic) and the cells of the other batches (pooled critic) are not detached and so receive gradient, while the barycenter support is a constant. No spectral normalisation is used.

Maximum mean discrepancy. Each batch is compared with the cells of all other batches in the minibatch, with the squared MMD as a biased (V-statistic) estimate,

$$\mathcal{A}_{\mathrm{MMD}} = \frac{1}{|K_{\mathcal{B}}|} \sum_{k \in K_{\mathcal{B}}} \widehat{\mathrm{MMD}}^2(\hat{P}_k, \hat{P}_{-k}), \qquad \widehat{\mathrm{MMD}}^2 = \frac{1}{n_k^2} \sum_{i,j \in \mathcal{B}_k} \kappa(u_i, u_j) + \frac{1}{n_{-k}^2} \sum_{i,j \in \mathcal{B}_{-k}} \kappa(u_i, u_j) - \frac{2}{n_k n_{-k}} \sum_{i \in \mathcal{B}_k} \sum_{j \in \mathcal{B}_{-k}} \kappa(u_i, u_j), \quad (9)$$

with $n_k = |\mathcal{B}_k|$, $n_{-k} = |\mathcal{B}_{-k}|$ and the multi-scale Gaussian kernel

$$\kappa(u, u') = \frac{1}{5} \sum_{m \in \{0.25,\, 0.5,\, 1,\, 2,\, 4\}} \exp\Big(- \frac{\|u - u'\|^2}{2\sigma_m^2} \Big), \qquad \sigma_m = \sqrt{d}\, m, \quad (10)$$

so that for $d = 10$ the bandwidths are $\sigma = 0.79$, 1.58, 3.16, 6.32 and 12.65. They are fixed and not tuned.

Sinkhorn divergence. With the Euclidean ground cost $c(u, u') = \|u - u'\|_2$ and entropic smoothing $\epsilon = 0.05\sqrt{2d}$,

$$\mathcal{A}_{\mathrm{SD}} = \frac{1}{|K_{\mathcal{B}}|} \sum_{k \in K_{\mathcal{B}}} S_\epsilon(\hat{P}_k, \hat{P}_{-k}), \qquad S_\epsilon(\alpha, \beta) = \mathrm{OT}_\epsilon(\alpha, \beta) - \tfrac{1}{2}\mathrm{OT}_\epsilon(\alpha, \alpha) - \tfrac{1}{2}\mathrm{OT}_\epsilon(\beta, \beta), \quad (11)$$

$$\mathrm{OT}_\epsilon(\alpha, \beta) = \max_{f, g} \langle \alpha, f \rangle + \langle \beta, g \rangle - \epsilon \left\langle \alpha \otimes \beta,\ e^{(f \oplus g - c)/\epsilon} - 1 \right\rangle, \quad (12)$$

which at the optimum equals $\langle \alpha, f \rangle + \langle \beta, g \rangle$. The potentials are found by Sinkhorn updates, annealed in ϵ from the diameter of the cost matrix with ratio 0.8 per step, then iterated at the final ϵ in double precision until the largest L_1 marginal error of the plans is below 10^{-4} (a cap of 500,000 iterations raises an error). The gradient is the envelope-theorem gradient: the potentials are held fixed and the cost carries gradient through its row point only. At this smoothing S_ϵ is below W_1 and is not an estimate of it.

scvi-tools adversarial classifier (batch-conditioned decoder only). A classifier with 32 hidden units, applied to a sample z_i from the posterior, is trained in one step per encoder step on $\kappa\, \mathcal{L}_{\mathrm{CE}}$ (as in Eq. 3, detached input), and the encoder minimises $\mathcal{L}_{\mathrm{scVI}} + \kappa\, \mathcal{A}_{\mathrm{JS}}$ with $\mathcal{A}_{\mathrm{JS}}$ as in Eq. 4, where $\kappa = 1 - \beta_e$ is fixed by the KL weight and there is no strength to vary.

Supplementary Table S6: Ranking detail: biology score at the matched batch removal b^* per penalty, task and decoder, with the leave-one-seed-out standard error (SE) of the reading (the same seed left out of plain scVI, b^* held fixed), the number m of the three leave-one-seed-out repeats that gave a reading, the rank (1 is the highest biology; ties share the average rank) and the lowest to highest rank over the three repeats. The SE and the rank range are given only where $m \geq 2$ (otherwise n/a; 0 of 78 ranked readings). "Cross." is the number of times the penalty's seed-mean curve crosses b^*; the reading is taken at the first crossing. In every pair every penalty's curve reaches b^*, so no reading is marked as not reached. 13 task and decoder pairs are ranked; the immune task and both simulations with the batch-conditioned decoder are low-signal and are in Part B. Simulations are ranked separately from the real tasks (Supplementary Notes 5 and 6; Table 2 gives the mean ranks). **Part B (last block)**: Low-signal task and decoder pairs (3; the gap between plain scVI's batch score and the highest batch score every penalty reaches is under 0.02): biology score at b^* read in the same way, not ranked and without standard errors (Supplementary Note 5).

Penalty	Biology at b^*	SE	m	Rank	Rank range	Cross.
Part A: ranked pairs						
Pancreas, batch-conditioned decoder; $b^* = 0.684$						
JS discriminator	0.691	0.001	3	6	6	2
Reference critic	0.697	0.005	3	4	3–5	2
Pooled critic	0.699	0.006	3	2	2–4	2
Barycenter critic	0.701	0.006	3	1	1–2	2
MMD	0.694	0.002	3	5	3–5	3
Sinkhorn	0.697	0.001	3	3	1–4	4
Lung, batch-conditioned decoder; $b^* = 0.648$						
JS discriminator	0.704	0.011	3	6	5–6	3
Reference critic	0.717	0.008	3	4	2–6	2
Pooled critic	0.719	0.003	3	3	3–4	2
Barycenter critic	0.715	0.005	3	5	3–5	2
MMD	0.723	0.003	3	2	1–3	2
Sinkhorn	0.726	0.003	3	1	1–2	4
ATAC small, batch-conditioned decoder; $b^* = 0.746$						
JS discriminator	0.585	0.003	3	6	6	2
Reference critic	0.603	0.003	3	5	5	2
Pooled critic	0.632	0.007	3	1	1–3	2
Barycenter critic	0.631	0.001	3	3	1–2	2
MMD	0.619	0.007	3	4	4	1
Sinkhorn	0.631	0.004	3	2	1–3	2
ATAC large, batch-conditioned decoder; $b^* = 0.632$						
JS discriminator	0.542	0.010	3	4	4–5	1
Reference critic	0.565	0.009	3	1	1–2	4
Pooled critic	0.546	0.014	3	3	2–5	2
Barycenter critic	0.563	0.004	3	2	1–3	2
MMD	0.536	0.017	3	6	3–6	1
Sinkhorn	0.540	0.017	3	5	3–6	4
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$						
JS discriminator	0.622	0.005	3	3	3–6	2
Reference critic	0.618	0.003	3	5	3–5	2
Pooled critic	0.624	0.002	3	2	2–3	2
Barycenter critic	0.619	0.004	3	4	4	2
MMD	0.617	0.009	3	6	2–6	2
Sinkhorn	0.630	0.002	3	1	1	2
Pancreas, unconditioned decoder; $b^* = 0.492$						
JS discriminator	0.585	0.008	3	2	1–5	1
Reference critic	0.582	0.004	3	3	2–4	1
Pooled critic	0.579	0.003	3	5	4–5	1
Barycenter critic	0.576	0.005	3	6	6	2
MMD	0.588	0.005	3	1	1–3	1
Sinkhorn	0.581	0.005	3	4	1–3	1
Lung, unconditioned decoder; $b^* = 0.535$						
JS discriminator	0.690	0.006	3	4	4–5	1
Reference critic	0.710	0.004	3	1	1	2
Pooled critic	0.703	0.007	3	3	2–3	1
Barycenter critic	0.690	0.005	3	5	4–5	1
MMD	0.707	0.001	3	2	2–3	1
Sinkhorn	0.685	0.004	3	6	6	1
Immune, unconditioned decoder; $b^* = 0.531$						
JS discriminator	0.720	0.012	3	1	1–3	1
Reference critic	0.716	0.002	3	2	1–3	1
Pooled critic	0.711	0.006	3	3	2–5	1
Barycenter critic	0.708	0.003	3	4	2–6	2
MMD	0.703	0.006	3	6	4–6	1
Sinkhorn	0.708	0.003	3	5	3–5	1

ATAC small, unconditioned decoder; $b^* = 0.522$						
JS discriminator	0.666	0.008	3	5	5–6	2
Reference critic	0.659	0.003	3	6	5–6	2
Pooled critic	0.677	0.004	3	4	3–4	2
Barycenter critic	0.700	0.015	3	1	1–2	2
MMD	0.692	0.015	3	2	1–4	1
Sinkhorn	0.684	0.013	3	3	1–3	1
ATAC large, unconditioned decoder; $b^* = 0.550$						
JS discriminator	0.617	0.009	3	4	2–5	1
Reference critic	0.621	0.006	3	2	2–4	2
Pooled critic	0.620	0.015	3	3	1–5	1
Barycenter critic	0.611	0.010	3	5	3–6	2
MMD	0.622	0.010	3	1	1–3	1
Sinkhorn	0.596	0.021	3	6	4–6	1
Human–mouse immune, unconditioned decoder; $b^* = 0.450$						
JS discriminator	0.617	0.001	3	5	2–4	1
Reference critic	0.589	0.030	3	6	6	2
Pooled critic	0.619	0.002	3	2	1–5	2
Barycenter critic	0.618	0.022	3	3	3–5	4
MMD	0.619	0.001	3	1	1–2	1
Sinkhorn	0.617	0.002	3	4	2–4	1
Simulation 1, unconditioned decoder; $b^* = 0.369$						
JS discriminator	0.709	0.002	3	1	1	1
Reference critic	0.691	0.003	3	2	2–3	3
Pooled critic	0.681	0.003	3	5	5	1
Barycenter critic	0.677	0.000	3	6	6	3
MMD	0.688	0.008	3	3	2–4	1
Sinkhorn	0.683	0.002	3	4	3–4	1
Simulation 2, unconditioned decoder; $b^* = 0.318$						
JS discriminator	0.584	0.002	3	1	1–3	2
Reference critic	0.579	0.002	3	4	3–5	3
Pooled critic	0.582	0.001	3	2	2	2
Barycenter critic	0.578	0.001	3	5	4–5	3
MMD	0.581	0.001	3	3	1–4	2
Sinkhorn	0.573	0.003	3	6	6	1
Part B: low-signal pairs (not ranked)						
Immune, batch-conditioned decoder; $b^* = 0.648$						
JS discriminator	0.772	n/a	–	not ranked	–	2
Reference critic	0.756	n/a	–	not ranked	–	2
Pooled critic	0.751	n/a	–	not ranked	–	2
Barycenter critic	0.749	n/a	–	not ranked	–	2
MMD	0.769	n/a	–	not ranked	–	2
Sinkhorn	0.766	n/a	–	not ranked	–	2
Simulation 1, batch-conditioned decoder; $b^* = 0.815$						
JS discriminator	0.897	n/a	–	not ranked	–	2
Reference critic	0.896	n/a	–	not ranked	–	2
Pooled critic	0.897	n/a	–	not ranked	–	2
Barycenter critic	0.896	n/a	–	not ranked	–	2
MMD	0.896	n/a	–	not ranked	–	4
Sinkhorn	0.897	n/a	–	not ranked	–	2
Simulation 2, batch-conditioned decoder; $b^* = 0.660$						
JS discriminator	0.493	n/a	–	not ranked	–	2
Reference critic	0.508	n/a	–	not ranked	–	2
Pooled critic	0.483	n/a	–	not ranked	–	2
Barycenter critic	0.458	n/a	–	not ranked	–	2
MMD	0.474	n/a	–	not ranked	–	2
Sinkhorn	0.494	n/a	–	not ranked	–	3

Supplementary Table S7: Placement with the unconditioned decoder (counterpart of Table 3 and of Figure 6). **(a)** Mean selection score ($0.4\times$batch $+ 0.6\times$biology) per method and real task at the strength (Harmony, Scanorama, sysVI: the setting) with the best mean selection score on real tasks of other task families (related tasks never choose for each other); in parentheses, the task's own best strength or setting (optimistic bound). Plain scVI here receives no batch information. The scvi-tools adversary is run with the batch-conditioned decoder only and is not shown. scANVI (uses cell-type labels), PCA and the three tools have no decoder, so their values equal those of Table 3. The last column counts the real tasks (of six) on which the method's score exceeds plain scVI's, at the transferred strength and, in parentheses, at the own best. **(b)** Number of real tasks on which the penalty's expected best selection score among k randomly drawn settings exceeds that of the tool at k settings of its own, $k = 1$ and $k = 6$; Scanorama at $k = 6$ is compared on the five tasks where all six of its settings exist (the 80-neighbour setting could not be fitted on the large chromatin-accessibility task). Counts are descriptive (Supplementary Notes 7 and 8).

(a)

Method	Pancreas	Lung	Immune	ATAC small	ATAC large	Human–mouse immune	Above plain scVI
Plain scVI	0.523	0.610	0.626	0.575	0.588	0.547	
JS discriminator	0.601 (0.654)	0.644 (0.644)	0.691 (0.691)	0.639 (0.639)	0.565 (0.594)	0.567 (0.575)	5 (6)
Reference critic	0.592 (0.650)	0.613 (0.651)	0.653 (0.671)	0.540 (0.645)	0.575 (0.607)	0.556 (0.580)	4 (6)
Pooled critic	0.579 (0.660)	0.644 (0.662)	0.648 (0.674)	0.591 (0.659)	0.589 (0.617)	0.564 (0.569)	6 (6)
Barycenter critic	0.560 (0.575)	0.607 (0.640)	0.636 (0.656)	0.572 (0.650)	0.447 (0.595)	0.533 (0.552)	2 (6)
MMD	0.600 (0.600)	0.636 (0.646)	0.663 (0.674)	0.634 (0.650)	0.609 (0.612)	0.558 (0.561)	6 (6)
Sinkhorn	0.575 (0.584)	0.621 (0.645)	0.655 (0.664)	0.533 (0.652)	0.515 (0.598)	0.531 (0.552)	3 (6)
Harmony	0.706 (0.710)	0.621 (0.621)	0.664 (0.664)	0.576 (0.587)	0.552 (0.558)	0.520 (0.531)	4 (4)
Scanorama	0.600 (0.613)	0.598 (0.600)	0.685 (0.692)	0.517 (0.538)	0.510 (0.522)	0.524 (0.525)	2 (2)
sysVI	0.723 (0.726)	0.649 (0.653)	0.685 (0.700)	0.533 (0.571)	0.511 (0.537)	0.501 (0.542)	3 (3)
scANVI (labels)	0.696	0.740	0.743	0.748	0.656	0.619	6
PCA (uncorrected)	0.577	0.585	0.565	0.490	0.527	0.478	1

(b)

	Plain scVI		Harmony		Scanorama		sysVI		scANVI (labels)		PCA	
Penalty	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6
JS discriminator	4/6	5/6	3/6	5/6	5/6	4/5	3/6	3/6	0/6	0/6	6/6	6/6
Reference critic	5/6	6/6	5/6	5/6	4/6	4/5	3/6	3/6	0/6	0/6	5/6	6/6
Pooled critic	5/6	6/6	4/6	5/6	5/6	4/5	3/6	4/6	0/6	0/6	6/6	6/6
Barycenter critic	1/6	6/6	2/6	4/6	3/6	3/5	3/6	3/6	0/6	0/6	5/6	5/6
MMD	5/6	6/6	4/6	5/6	4/6	3/5	3/6	3/6	0/6	0/6	5/6	6/6
Sinkhorn	4/6	6/6	4/6	4/6	4/6	3/5	3/6	3/6	0/6	0/6	5/6	6/6

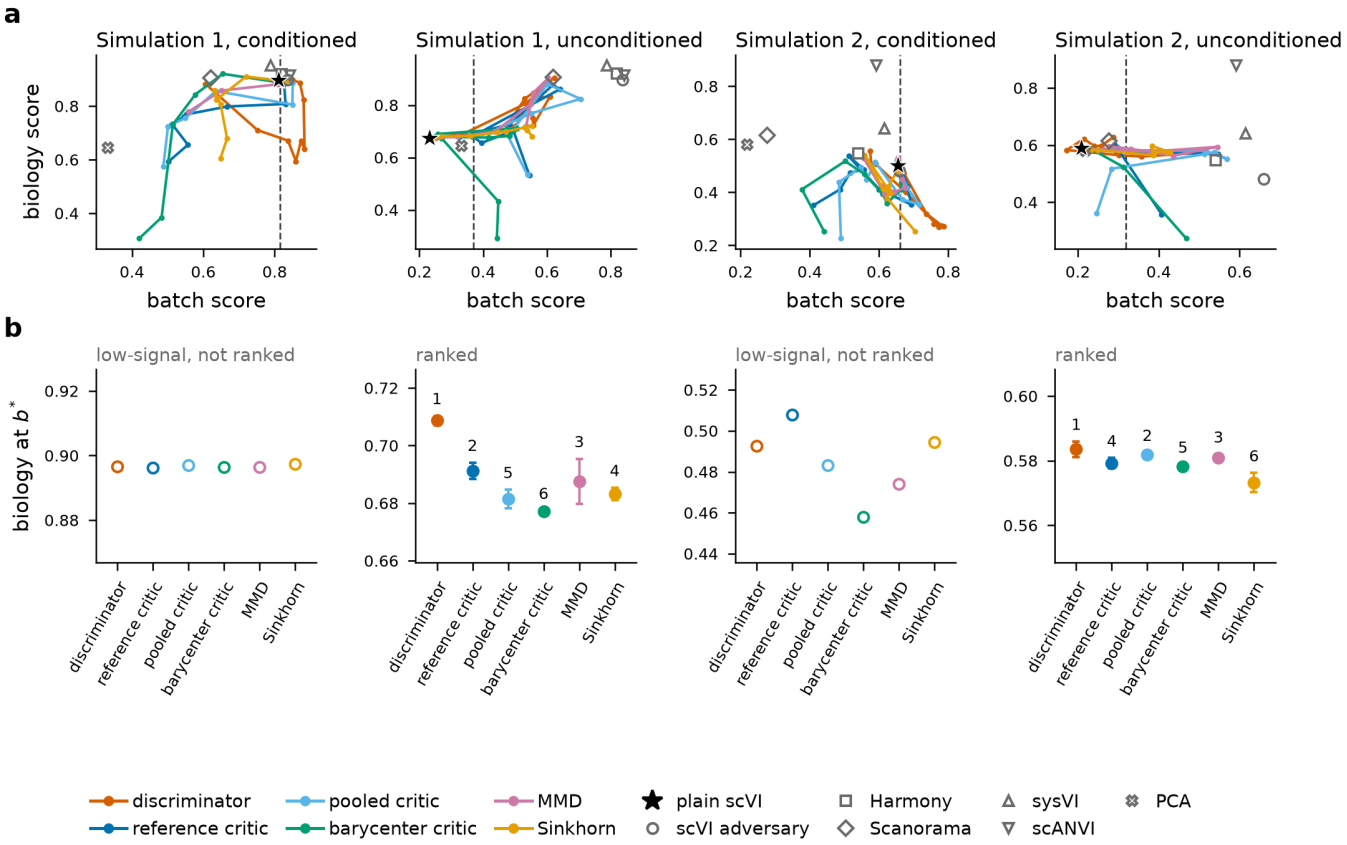
Supplementary Table S9: Replication across training seeds: biology at the matched batch removal b^* (fixed at its sweep value) on new seeds 0–4 beside the sweep reading (seeds 100–102), per penalty, task and decoder, for the 13 ranked pairs. "Strengths" are the two bracketing strengths (0 is plain scVI at seeds 0–4); "Batch" and "Biology" give the seed-mean scores at those two points; the reading interpolates linearly in batch score between them at the first crossing of b^*, walking up from plain scVI at seeds 0–4 (in 7 readings the new-seed mean at the lower refitted strength already reached b^*, so the reading starts at plain scVI). Rank: replication (sweep); 1 is the highest biology; a penalty that does not reach b^* takes the bottom place. "extension": the new-seed means did not bracket b^* at the planned strengths, so the next sweep strength was fitted once with seeds 0–4, as the agreement rule prescribes (8 readings, 40 fits). †Did not replicate the crossing: the highest new-seed mean batch score is shown. Each block header gives b^*, plain scVI's batch score at seeds 0–4 beside the sweep's and Kendall's τ_b between the sweep and replication ranks. All 895 fits were scored and 0 failed. The last block gives the four large sweep gaps (0.04 or more within a pair) and their sign on the new seeds (Supplementary Notes 4 and 10).

Penalty	Strengths	Batch	Biology	At b^*, new	At b^*, sweep	Rank	Note
Pancreas, batch-conditioned decoder; b^* = 0.684; plain scVI batch score 0.668 (sweep 0.672); τ_b = −0.067							
JS discriminator	0.3–1	0.681/0.703	0.697/0.688	0.695	0.691	4 (6)	
Reference critic	0–0.3	0.668/0.685	0.696/0.704	0.703	0.697	1 (4)	
Pooled critic	0.3–1	0.677/0.696	0.698/0.694	0.696	0.699	3 (2)	
Barycenter critic	1–3	0.679/0.695	0.697/0.691	0.695	0.701	5 (1)	
MMD	up to 3	0.674 (max)	–	–	0.694	6 (5)	extension; not reached†
Sinkhorn	3–10	0.682/0.696	0.699/0.701	0.699	0.697	2 (3)	extension
Pancreas, unconditioned decoder; b^* = 0.492; plain scVI batch score 0.451 (sweep 0.446); τ_b = 0.200							
JS discriminator	0.1–0.3	0.488/0.552	0.587/0.614	0.589	0.585	1 (2)	
Reference critic	1–3	0.481/0.517	0.582/0.594	0.585	0.582	2 (3)	
Pooled critic	1–3	0.492/0.536	0.584/0.599	0.584	0.579	3 (5)	extension
Barycenter critic	1–3	0.481/0.497	0.580/0.586	0.584	0.576	4 (6)	
MMD	10–30	0.480/0.527	0.578/0.589	0.581	0.588	5 (1)	
Sinkhorn	3–10	0.490/0.495	0.580/0.580	0.580	0.581	6 (4)	
Lung, batch-conditioned decoder; b^* = 0.648; plain scVI batch score 0.633 (sweep 0.633); τ_b = 0.867							
JS discriminator	0.3–1	0.644/0.671	0.713/0.610	0.700	0.704	6 (6)	
Reference critic	0.3–1	0.647/0.661	0.723/0.707	0.722	0.717	4 (4)	
Pooled critic	0.3–1	0.647/0.663	0.726/0.708	0.725	0.719	2 (3)	
Barycenter critic	0–1	0.633/0.649	0.723/0.720	0.720	0.715	5 (5)	
MMD	3–10	0.640/0.650	0.728/0.722	0.723	0.723	3 (2)	
Sinkhorn	3–10	0.646/0.653	0.730/0.713	0.726	0.726	1 (1)	
Lung, unconditioned decoder; b^* = 0.535; plain scVI batch score 0.479 (sweep 0.481); τ_b = 0.600							
JS discriminator	0.1–0.3	0.531/0.565	0.687/0.683	0.687	0.690	5 (4)	
Reference critic	1–3	0.505/0.539	0.697/0.705	0.704	0.710	2 (1)	
Pooled critic	1–3	0.519/0.557	0.701/0.698	0.700	0.703	3 (3)	
Barycenter critic	100–300	0.523/0.557	0.689/0.670	0.682	0.690	6 (5)	
MMD	10–30	0.511/0.550	0.707/0.710	0.709	0.707	1 (2)	
Sinkhorn	30–100	0.521/0.612	0.705/0.660	0.698	0.685	4 (6)	
Immune, unconditioned decoder; b^* = 0.531; plain scVI batch score 0.499 (sweep 0.499); τ_b = 0.200							
JS discriminator	0.03–0.1	0.505/0.531	0.704/0.720	0.720	0.720	1 (1)	
Reference critic	0.3–1	0.513/0.537	0.716/0.706	0.709	0.716	6 (2)	
Pooled critic	0.3–1	0.510/0.536	0.712/0.718	0.717	0.711	3 (3)	
Barycenter critic	0–3	0.499/0.534	0.711/0.715	0.715	0.708	4 (4)	
MMD	10–30	0.525/0.563	0.711/0.710	0.711	0.703	5 (6)	
Sinkhorn	10–30	0.528/0.543	0.717/0.719	0.718	0.708	2 (5)	
ATAC small, batch-conditioned decoder; b^* = 0.746; plain scVI batch score 0.690 (sweep 0.694); τ_b = 0.467							
JS discriminator	0.3–1	0.719/0.753	0.628/0.576	0.587	0.585	6 (6)	
Reference critic	0.3–1	0.720/0.773	0.638/0.593	0.616	0.603	5 (5)	
Pooled critic	0.3–1	0.724/0.779	0.634/0.601	0.621	0.632	4 (1)	
Barycenter critic	1–3	0.738/0.772	0.648/0.615	0.640	0.631	1 (3)	
MMD	10–30	0.725/0.757	0.632/0.623	0.626	0.619	3 (4)	
Sinkhorn	3–10	0.732/0.761	0.639/0.620	0.630	0.631	2 (2)	
ATAC small, unconditioned decoder; b^* = 0.522; plain scVI batch score 0.398 (sweep 0.402); τ_b = 0.600							
JS discriminator	0.1–0.3	0.460/0.617	0.682/0.650	0.669	0.666	5 (5)	
Reference critic	1–3	0.496/0.678	0.678/0.589	0.665	0.659	6 (6)	
Pooled critic	1–3	0.509/0.682	0.682/0.631	0.678	0.677	3 (4)	
Barycenter critic	3–10	0.520/0.619	0.677/0.690	0.677	0.700	4 (1)	extension
MMD	10–30	0.438/0.552	0.683/0.692	0.689	0.692	1 (2)	
Sinkhorn	10–30	0.521/0.596	0.685/0.665	0.685	0.684	2 (3)	extension
ATAC large, batch-conditioned decoder; b^* = 0.632; plain scVI batch score 0.591 (sweep 0.589); τ_b = 0.600							
JS discriminator	0.1–0.3	0.621/0.661	0.525/0.502	0.518	0.542	6 (4)	
Reference critic	0.1–0.3	0.618/0.663	0.548/0.552	0.550	0.565	1 (1)	
Pooled critic	0.1–0.3	0.614/0.654	0.540/0.550	0.545	0.546	3 (3)	
Barycenter critic	0.3–1	0.620/0.636	0.550/0.544	0.546	0.563	2 (2)	
MMD	3–10	0.612/0.652	0.536/0.543	0.540	0.536	4 (6)	
Sinkhorn	0–10	0.591/0.633	0.553/0.537	0.538	0.540	5 (5)	

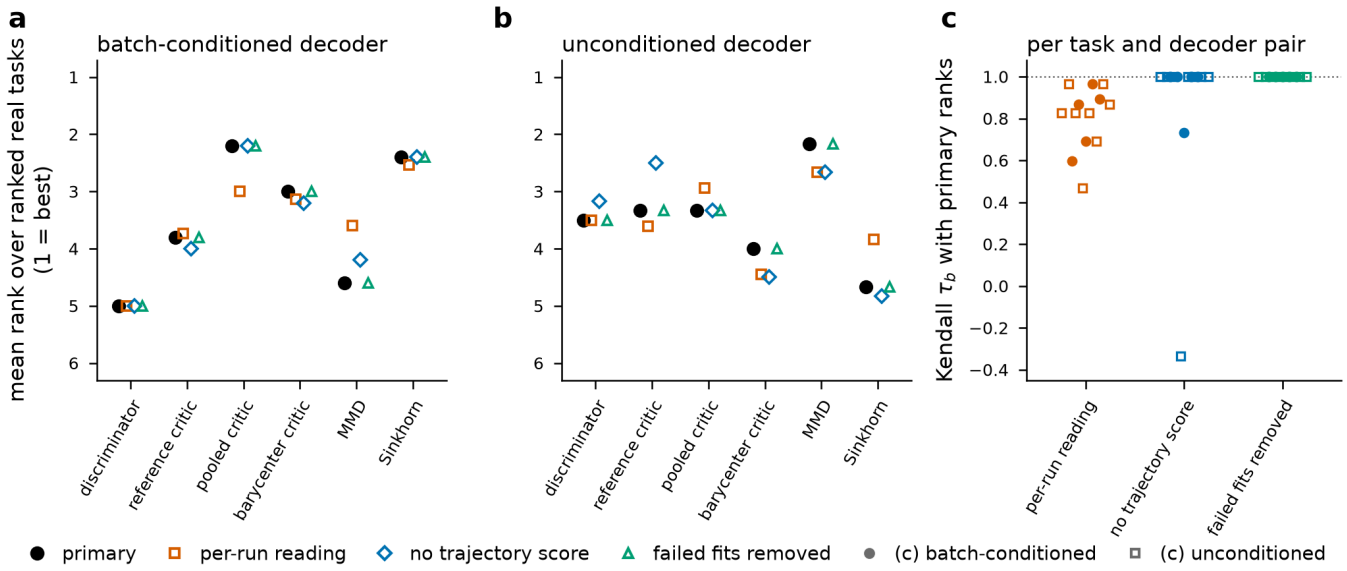
ATAC large, unconditioned decoder; $b^* = 0.550$; plain scVI batch score 0.538 (sweep 0.538); $\tau_b = 0.467$							
JS discriminator	0.03–0.1	0.543/0.587	0.601/0.588	0.599	0.617	5 (4)	
Reference critic	0.1–0.3	0.544/0.582	0.628/0.586	0.622	0.621	1 (2)	
Pooled critic	0.1–0.3	0.544/0.558	0.635/0.604	0.622	0.620	2 (3)	
Barycenter critic	0–1	0.538/0.554	0.637/0.612	0.619	0.611	3 (5)	
MMD	3–10	0.542/0.553	0.629/0.603	0.611	0.622	4 (1)	
Sinkhorn	30–100	0.543/0.565	0.616/0.522	0.587	0.596	6 (6)	
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$; plain scVI batch score 0.540 (sweep 0.540); $\tau_b = -0.067$							
JS discriminator	0–0.1	0.540/0.557	0.630/0.613	0.617	0.622	4 (3)	
Reference critic	0.3–1	0.553/0.570	0.620/0.590	0.619	0.618	2 (5)	
Pooled critic	0.1–0.3	0.547/0.556	0.630/0.598	0.606	0.624	6 (2)	
Barycenter critic	1–3	0.552/0.557	0.619/0.606	0.615	0.619	5 (4)	extension
MMD	3–10	0.549/0.562	0.622/0.606	0.617	0.617	3 (6)	
Sinkhorn	3–10	0.553/0.559	0.624/0.616	0.624	0.630	1 (1)	
Human–mouse immune, unconditioned decoder; $b^* = 0.450$; plain scVI batch score 0.432 (sweep 0.432); $\tau_b = 0.333$							
JS discriminator	0.01–0.03	0.436/0.451	0.608/0.617	0.617	0.617	1 (5)	
Reference critic	0–0.3	0.432/0.451	0.610/0.603	0.603	0.589	6 (6)	
Pooled critic	0–0.3	0.432/0.457	0.610/0.619	0.616	0.619	2 (2)	
Barycenter critic	10–30	0.444/0.457	0.621/0.610	0.616	0.618	4 (3)	
MMD	3–10	0.445/0.485	0.619/0.598	0.616	0.619	3 (1)	
Sinkhorn	3–10	0.450/0.453	0.615/0.615	0.615	0.617	5 (4)	
Simulation 1, unconditioned decoder; $b^* = 0.369$; plain scVI batch score 0.232 (sweep 0.232); $\tau_b = 0.867$							
JS discriminator	0.1–0.3	0.338/0.557	0.681/0.832	0.703	0.709	1 (1)	
Reference critic	1–3	0.338/0.484	0.679/0.721	0.688	0.691	3 (2)	
Pooled critic	1–3	0.360/0.487	0.679/0.713	0.682	0.681	5 (5)	extension
Barycenter critic	1–3	0.285/0.403	0.677/0.679	0.678	0.677	6 (6)	
MMD	10–30	0.321/0.449	0.692/0.694	0.693	0.688	2 (3)	
Sinkhorn	3–10	0.336/0.424	0.684/0.695	0.688	0.683	4 (4)	
Simulation 2, unconditioned decoder; $b^* = 0.318$; plain scVI batch score 0.203 (sweep 0.209); $\tau_b = 0.867$							
JS discriminator	0.1–0.3	0.222/0.423	0.586/0.586	0.586	0.584	1 (1)	
Reference critic	3–10	0.284/0.461	0.581/0.573	0.580	0.579	3 (4)	
Pooled critic	3–10	0.305/0.505	0.581/0.574	0.581	0.582	2 (2)	
Barycenter critic	3–10	0.255/0.332	0.582/0.576	0.577	0.578	5 (5)	
MMD	10–30	0.254/0.346	0.584/0.578	0.580	0.581	4 (3)	
Sinkhorn	30–100	0.307/0.432	0.574/0.603	0.576	0.573	6 (6)	extension

Large sweep gaps

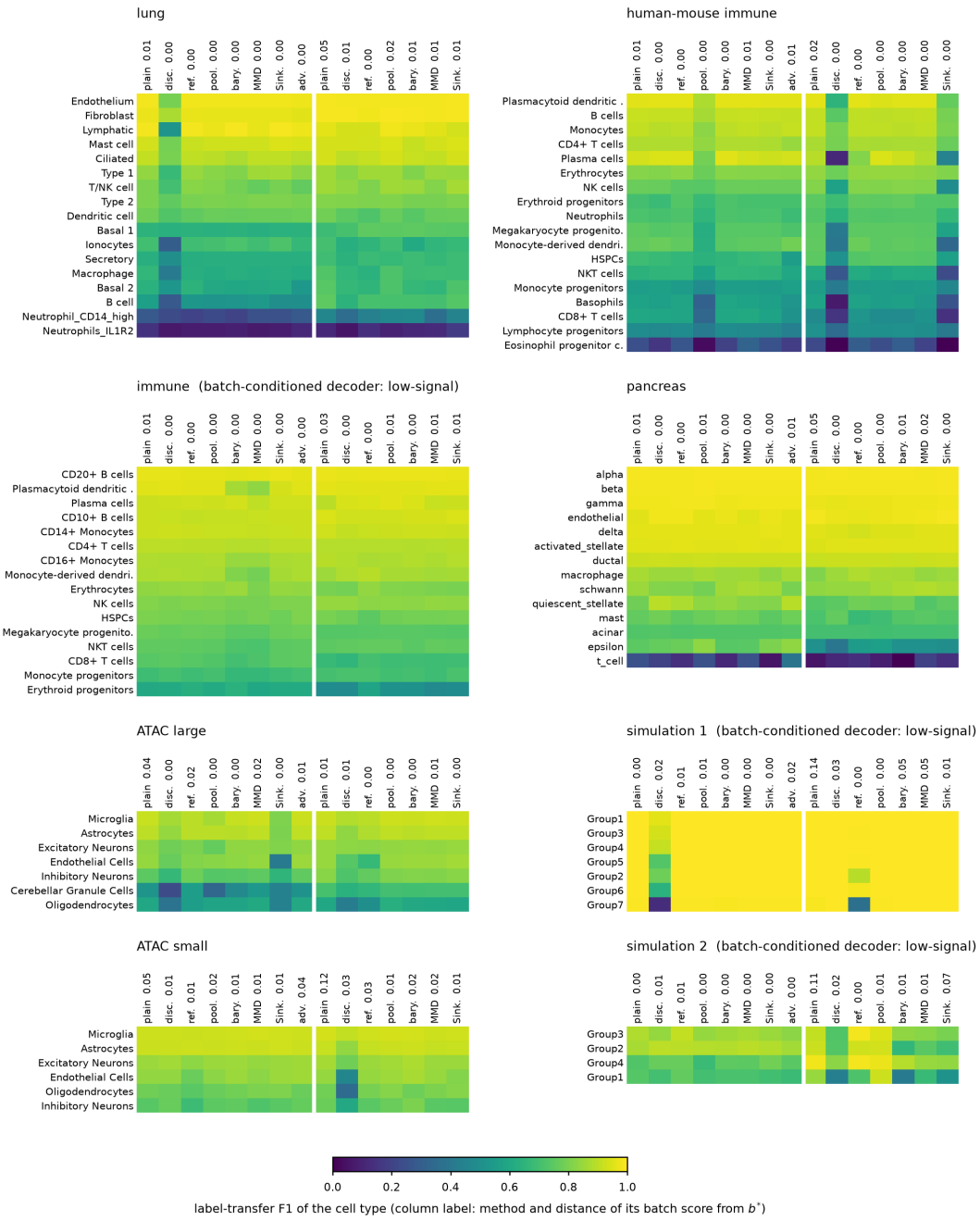
Pair	Difference	Sweep	New	Sign kept
ATAC small, unconditioned decoder	Barycenter critic − Reference critic	+0.041	+0.013	yes
ATAC small, batch-conditioned decoder	Barycenter critic − JS discriminator	+0.046	+0.053	yes
ATAC small, batch-conditioned decoder	JS discriminator − Pooled critic	−0.046	−0.034	yes
ATAC small, batch-conditioned decoder	JS discriminator − Sinkhorn	−0.046	−0.043	yes



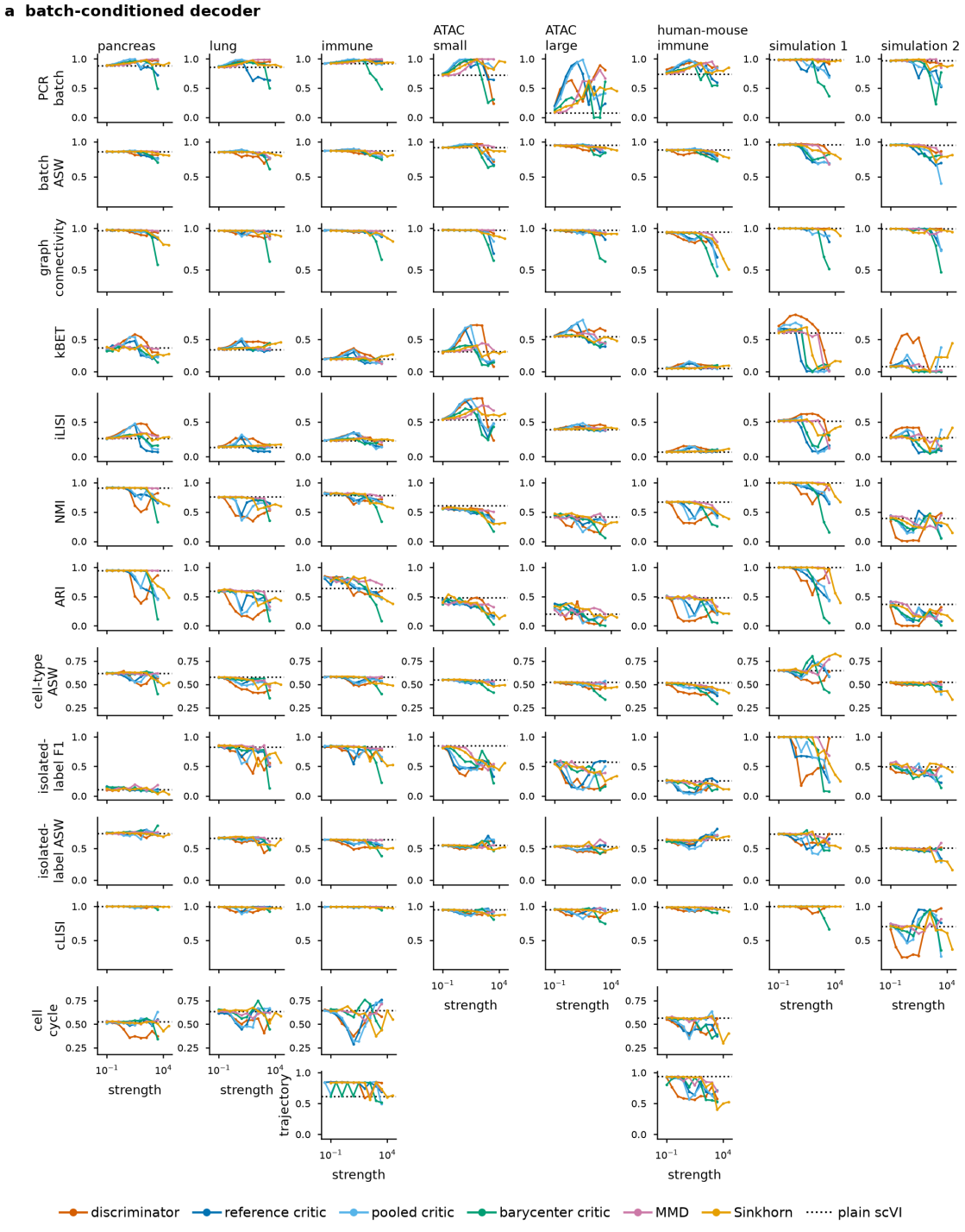
Supplementary Figure S1: Simulation tasks, each on its own, both decoders. **(a)** Batch score against biology score for each penalty's sweep strengths (seed means over seeds 100–102), plain scVI (star), the matched batch removal b^* (dashed line) and the best setting of each standard tool by selection score (open grey markers). **(b)** Biology score at b^* per penalty with its leave-one-seed-out standard error, and the rank above each point. With the batch-conditioned decoder both simulations are low-signal: their readings are shown with open markers, without standard errors and ranks. Simulations are ranked separately from the real tasks (Supplementary Notes 5 and 6; Supplementary Table S6).



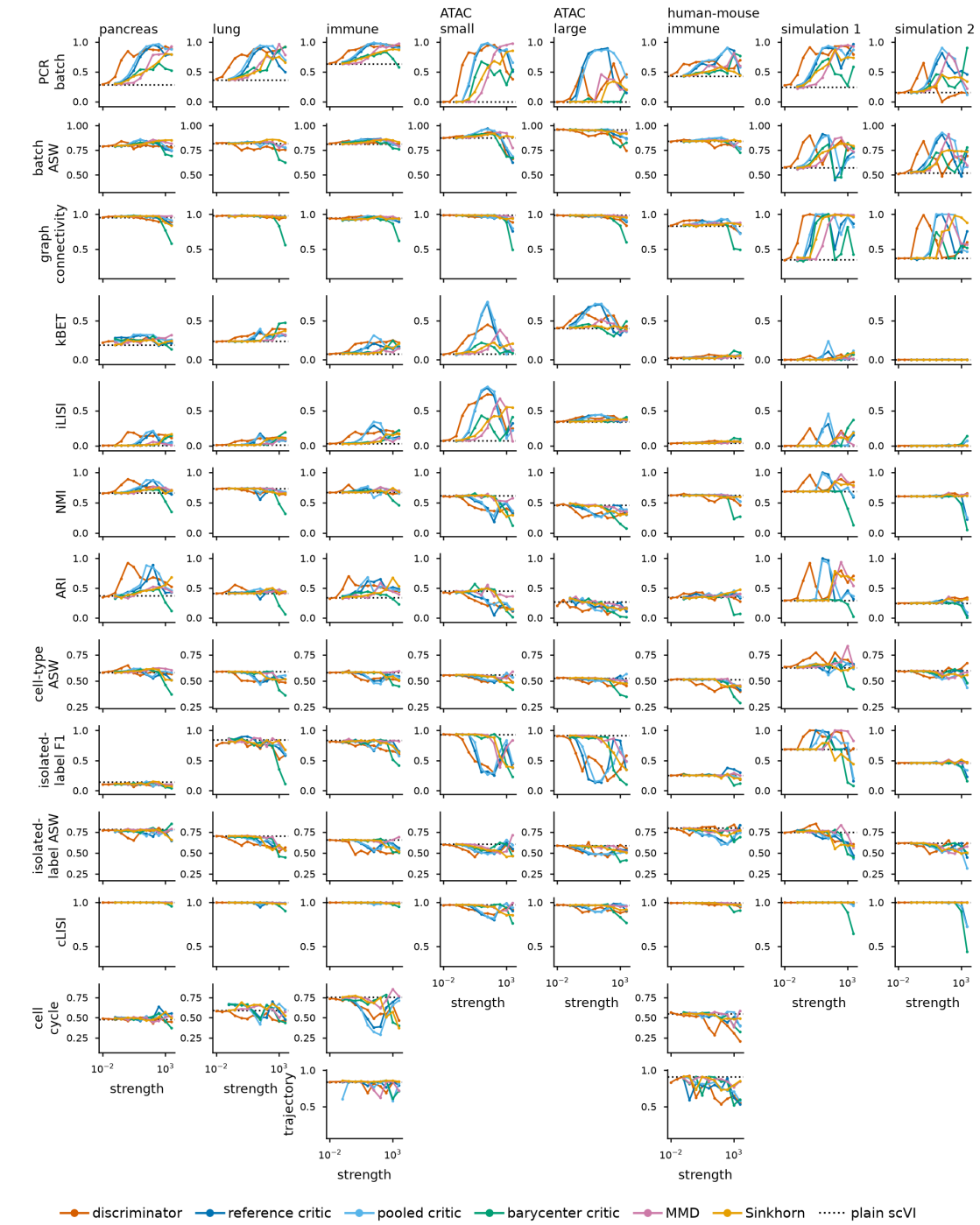
Supplementary Figure S2: Sensitivity of the ranking to the reading rules. Each variant re-reads biology at b^* from the same sweep fits: the per-run reading (each seed read on its own curve and the three per-seed ranks averaged), biology without trajectory conservation on the two immune tasks, and the primary reading with points that contain a failed fit removed (the sweep has no failed fit, so this variant reproduces the primary ranks). **(a, b)** Mean rank over the ranked real tasks per penalty and decoder. **(c)** Kendall's τ_b between the primary ranks of the six penalties and each variant's ranks in each of the 13 ranked task and decoder pairs (simulations included); points at 1 overlap. Descriptive; Supplementary Notes 5 and 6.



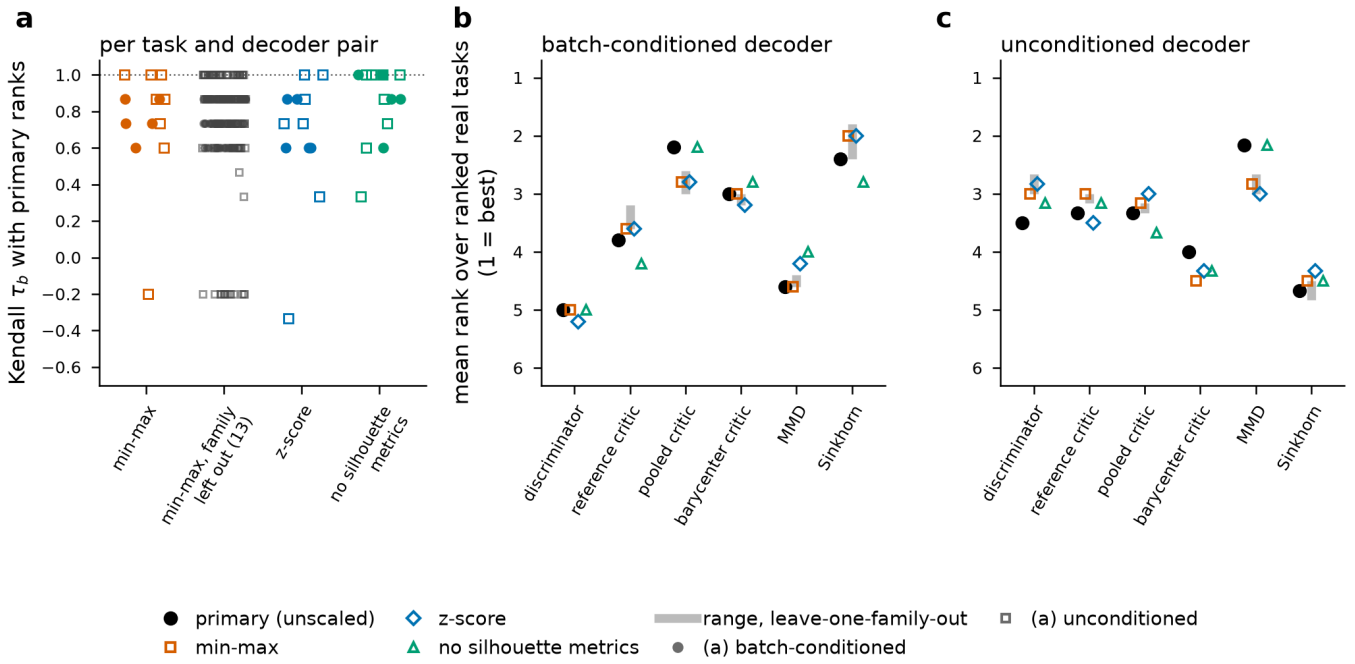
Supplementary Figure S3: Label transfer: F1 per cell type, method and task at the matched strength. Left block of each panel: batch-conditioned decoder (plain scVI, the six penalties and the scvi-tools adversary); right block: unconditioned decoder (plain scVI, which receives no batch information, and the six penalties). Each batch is held out in turn and its cells are predicted from the other batches by a 15-nearest-neighbour classifier; cell types present in one batch only are dropped; a type's F1 is averaged over the held-out batches that contain it and over seeds (three for plain scVI and the penalties, five for the scvi-tools adversary). Penalties are at the strength whose seed-mean batch score is nearest the matched batch removal b^*, never chosen by accuracy; plain scVI and the scvi-tools adversary are unmatched. The number in each column label is the distance between the column's seed-mean batch score and b^*. Cell types are ordered by their mean F1 over the columns of the panel. The immune task and both simulations with the batch-conditioned decoder are low-signal; they are tabulated here and left out of summaries (Supplementary Note 9).



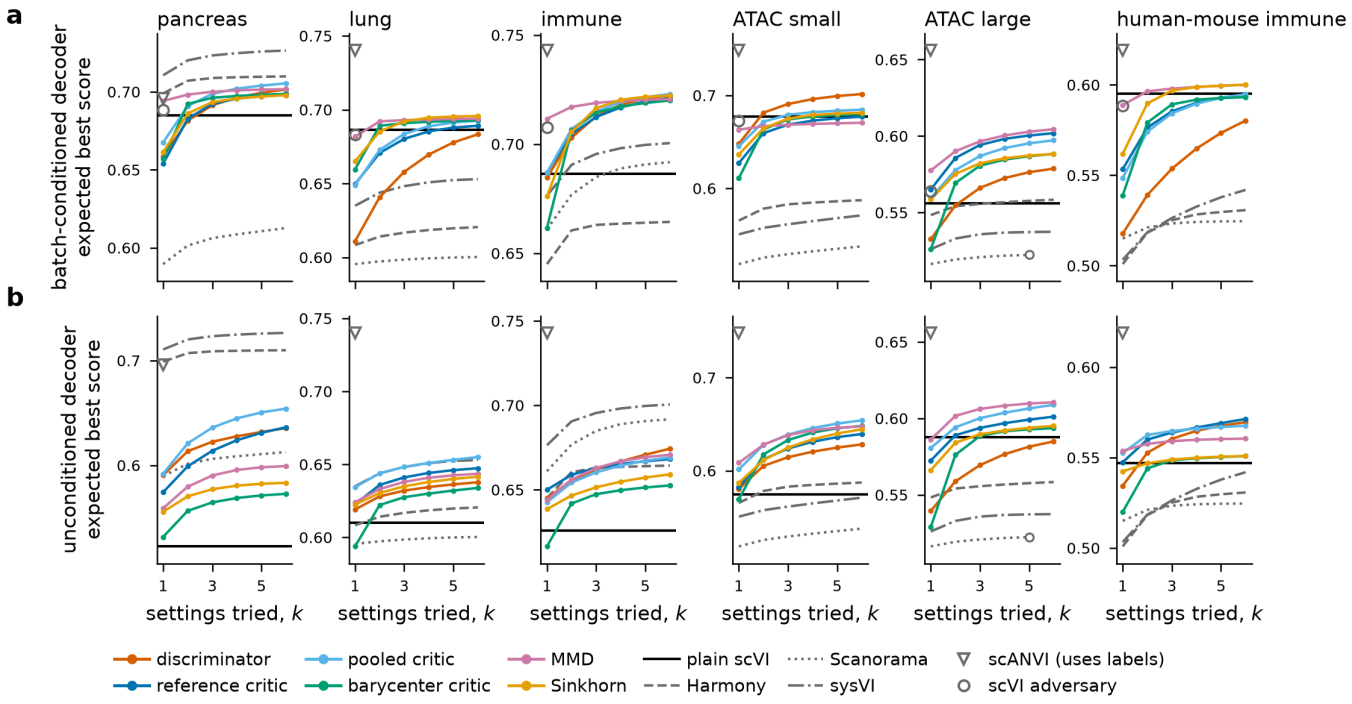
Supplementary Figure S4a: Per-metric strength curves, batch-conditioned decoder. Each batch metric (rows 1–5) and biology metric (rows 6–13) of the scIB set, unscaled, against penalty strength on a log scale, seed means over the three sweep seeds, one line per penalty, per task (columns). The dotted line is plain scVI. Each row has one vertical range for all tasks. Cell-cycle conservation exists for the four RNA real tasks and trajectory conservation for the two immune tasks; other panels are empty (Supplementary Note 3).



Supplementary Figure S4b: Per-metric strength curves, unconditioned decoder. As Supplementary Figure S4a, with the unconditioned decoder.



Supplementary Figure S5: Scaling checks (exploratory, used for no conclusion). The six penalties are re-ranked at matched batch removal after changing how the metrics are scaled or combined: scIB min-max scaling of each metric over one comparison set per task (every scored fit of the task, both decoders, all penalty strengths, plain scVI and the standard tools); the same with each of the 13 method families left out of the set that fixes the minimum and maximum; z-score standardisation over the same set; and a composite without the three silhouette metrics (batch average silhouette width, cell-type average silhouette width, isolated-label silhouette). The primary reading uses unscaled metrics. Batch and biology scores, b^*, the readings and the ranks are recomputed for every variant on the 13 ranked task and decoder pairs. **(a)** Kendall's τ_b between the primary ranks and each variant's ranks, one point per pair (leave-one-family-out: one point per pair and family). **(b, c)** Mean rank over the ranked real tasks per penalty and decoder; the grey bar is the range over the 13 leave-one-family-out runs.



Supplementary Figure S6: Best of k settings per task. Expected best selection score ($0.4\times$batch $+ 0.6\times$biology) among k settings drawn at random, $k = 1$ to 6, for each penalty, Harmony, Scanorama and sysVI, one panel per real task and decoder. Plain scVI (black line), scANVI (uses cell-type labels) and the scvi-tools adversary (batch-conditioned decoder only) have one setting. Scanorama at $k = 6$ is missing on the large chromatin-accessibility task (open marker at $k = 5$), where its 80-neighbour setting could not be fitted. The mean of the batch-conditioned curves over the six real tasks is Figure 6a. Every value is optimistic, because the best setting is chosen and scored on the same data (Supplementary Note 7).